I-FEST² Program Research Report

Structural Design and Analysis Optimization of Multi-scene Modular Interconnected Robotic Arm

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Abstract

With the continuous increase in the level of industrial automation and the demand in the field of medical rescue, there is a growing need for robotic arm systems that are high-precision, high-speed, and highly reliable. Traditional rigid serial robotic arms have issues such as large size, limited usage scenarios, and poor ability to pass through complex narrow environments, as their fixed structures cannot adapt to different task scenarios. By following a modular design concept, a multi-scenario modular serial robotic arm has been developed. This arm features a modular mechanical structure based on standard interfaces of tenon and mortise joints, completing the establishment of a hardware and software platform and initial functional experimental verification. A kinematic model of the robotic arm has been built based on coordinate transformation method, using MATLAB to solve forward and inverse kinematic equations. Static stress finite element analysis, simulation analysis, modular combination analysis, and redundant degree of freedom optimization have been conducted for single-joint and multi-joint robotic arms. Pre-processing of the robotic arm structure assembly has been done from the perspectives of system energy consumption and workspace, enabling real-time optimization from "fixed demand mapping fixed structure" to "different demand mapping different structure" through modular recombination based on different scenarios. A multi-scenario switching and modular servo control system has been established, achieving high flexibility and versatility of the multi-scenario modular serial robotic arm, making it easy to maintain, expand, and optimize spatially. This development meets the needs of technological advancement and evolving requirements, ultimately achieving a harmonious integration of technology and innovation.

Keywords: Serial robotic arm, Modular, Structural design, Kinematic analysis, Spatial optimization

Table of Contents

Abstract I
Table of Contents II
1. Introduction 1
1.1 Background and Significance of the Research Project 1
1.2 Research Status of Modular Robotic Arms 2
1.3 Research Ideas and Methods of Multi-scene Modular Robotic Arms 3
1.3.1 Research Ideas of Multi-scenario Modular Structure 3
1.3.2 Research Methodology 4
1.3.3 Summary of the Author's Learning and Research Experience 6
2. Multi-scene Modular Series Manipulator Structure Design 7
2.1 Working Principle and Layout Design 7
2.1.1 Analysis of Working Principle 7
2.1.2 Modular Layout Design 8
2.2 Modular Connection Principle and Design 8
2.2.1 Modular Connection Principle 8
2.2.2 Modular Connection Design 10
2.3 Modular Connection Principle and Design 14
2.3.1 Selection and Modeling of the Driving Device 14
2.3.2 Design of Control System 15
2.3.3 Design of End Effectors 18
2.3.4 Overall Structural Assembly 20
3. Analysis and Optimization of Multi-scenario Modular Tandem Manipulators 21
3.1 Finite Element Analysis of Static Stress of Manipulator 21
3.1.1 Selection and Modeling of the Drive Device 21
3.1.2 Static Analysis of Each Joint and Overall of the Robotic Arm 22
3.1.3 Conclusion of Static Analysis 29
3.2 Analysis of Modular Assembly of Robotic Arms 30
3.2.1 Overview of Modular Assembly Analysis 30
3.2.2 Establish a World Coordinate System 30
3.2.3 Define the Pose of Each Module 30
3.2.4 Calculate the Total Number of Analysis Assembly Types 30
3.2.5 Conclusion of Assembly Analysis 34
3.3 Kinematic Modeling of Robotic Arms 34
3.3.1 Establishment of Linkage Coordinate System 34
3.3.2 Kinematic Equations of the Robotic Arm 35
3.3.3 Robotic Arm Modeling by MATLAB 36
3.4 Robotic Arm Redundant Degrees of Freedom 36
3.4.1 Analysis of Redundant Degrees of Freedom of Robotic Arm 36
3.4.2 Kinematic Solution of Robotic Arm 37
3.4.3 Robotic Arm Trajectory Planning 39
3.4.4 Kinematic Simulation of Robotic Arm 40
3.4.5 Optimization of Energy Consumption of Redundant Degrees of Freedom of the

Robotic Arm 44
3.4.6 Optimization of Redundant Degrees of Freedom Space of the Robotic Arm 45
4. Functional Verification of Multi-scene Modular Series Manipulator 47
4.1 Overview of Basic Functional Verification 47
4.2 Structural Design Verification 47
4.2.1 Functional Verification of Modular Connection Components 47
4.2.2 Hardware Structure Installation and Testing 48
4.2.3 Construction and Debugging of Software Platform 49
4.3 Analysis, Optimization and Verification 49
4.3.1 Kinematic Solution and Trajectory Planning Entity Verification 49
4.3.2 Verification of Redundant Degrees of Freedom Space Optimization Entities 49
5. Prototype Experimental Platform and Analysis Method Update and Iteration 50
5.1 Follow-up Research and Update Iterative Overview 50
5.2 Structural Design 50
5.2.1 Modular Linkages 50
5.2.2 Modular Joints 52
5.2.3 Working Platform 52
5.3 Analysis and Optimization 54
5.3.1 Modular Combination Analysis 54
5.3.2 Modular Reorganization and Optimization 55
6. Conclusions and Prospects 57
Acknowledgements 59
References 61
Appendix 62
Research Results Obtained 62

1. Introduction

1.1 Background and Significance of the Research Project

In today's world, the vast majority of robots are designed to increase the productivity of technicians and workers, and they are used to replace humans in monotonous, repetitive or dangerous tasks. These robots have a wide range of applications in many fields and have important research value. In particular, robots that mimic the structure of human upper limbs, they are widely used and belong to the discrete type of robots. This type of robot consists of joints with a single degree of freedom and rigid linkages, whose operation depends on the movement of the joints in each degree of freedom, and rigid linkages are used to maintain the geometry between the joints. With the drive control of a single-degree-of-freedom joint, these robots are able to flexibly change the position and attitude of the end effector [1].

Traditional discrete robots can be regarded as discrete joint combinations connected by fixed and rigid structures, but due to the limitations of their structure, such robots lack the ability to adapt to complex environments and spatial changes, and it is difficult to meet the needs of different tasks in various scenarios.



Figure 1-1 Harbin Institute of Technology wire-driven charging machine[3]

In contrast, some creatures in nature, such as the tentacles and trunks of the octopus, have a flexible body structure that allows for excellent supple handling and

precise positioning of objects. The morphological characteristics of these organisms provide inspiration for the design and control of continuum robots. Continuous robots are bionic robots made of flexible materials, without any discrete joints, with theoretically unlimited degrees of freedom, capable of performing complex grasping and positioning tasks, and excelling in environmental and spatial adaptability. However, the complex and fixed flexible structure of continuous robots makes them difficult to maintain and their scalability is limited [2].

In industrial automation production, it is necessary to determine the fixed robotic arm structure according to the actual task requirements and working scenarios, and one structure can only correspond to one demand, and it needs to be redesigned after iterating the production mode. In the field of medical rescue, robotic arms have to face complex and changeable working spaces, so there are high requirements for the robustness and spatial adaptability of robotic arms.

1.2 Research Status of Modular Robotic Arms

For the problem of poor environmental adaptability and versatility of robotic arms, some science and technology enterprises and research institutes have carried out research on modular robotic arms. As shown in Figure 1-2, the team of Harbin Institute of Technology has developed a new modular rotary joint manipulator that separates the drive device from the modular connection assembly. The motor, encoder, and brake axes are arranged in parallel, with standard mechanical and electrical interfaces, and can be assembled into different configurations of robotic arms according to requirements [4].

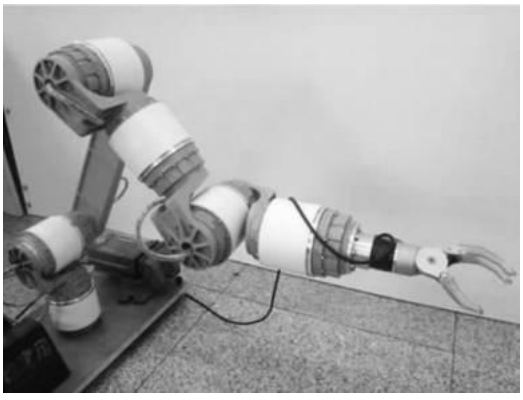


Figure 1-2 Harbin Institute of Technology modular rotary joint robotic arm

1.3 Research Ideas and Methods of Multi-scene Modular Robotic Arms

1.3.1 Research Ideas of Multi-scenario Modular Structure

Based on the flexible and flexible structure of the continuous manipulator with multiple degrees of freedom, on the basis of the rigid rigid discrete tandem manipulator rigid structure, as shown in Figure 1-3, the fixed rigid connecting rod is split into two modular interfaces with variable angles between adjacent joint shafts (the generalized connecting rod parameter is defined as α), which increases the degree of freedom of the manipulator, and designs a standardized mechanical mechanism structure and hardware electrical connection components of the module connection node, so that the number of joints and the degree of freedom can be flexibly adjusted. The robotic arm with a discrete fixed mechanism is modularly adjusted, and its movement after classification and assembly tends to be more than that of the non-discrete continuous robotic arm, which increases the flexibility and scalability of the robotic arm. As shown in Figure 1-3, a variety of expansion interfaces are designed on the end-effector to connect sensor modules suitable for various scenarios.

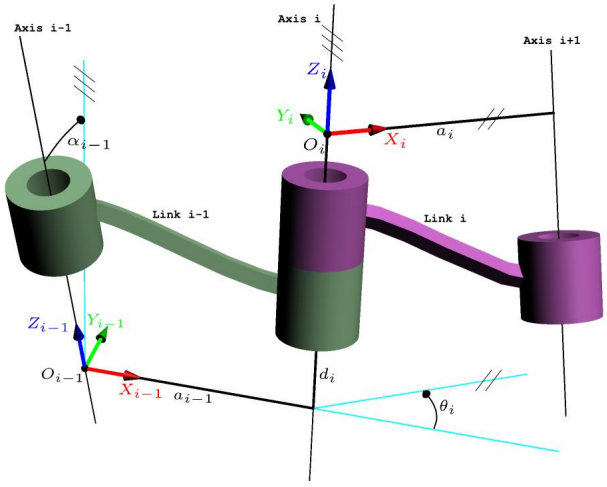


Figure 1-3 SDH modeling method of robot linkage coordinate system

The robotic arm is no longer limited to the grasping work of the manipulator, and can further realize the detection and analysis of different scenarios, enhance the

versatility and environmental adaptability of the robotic arm, and meet the needs of industrial automation manufacturing, industrial flaw detection, medical equipment, disaster rescue and other work. The overall modular design enables each joint and tool module of the robotic arm to be flexibly disassembled and repaired separately, reducing the maintenance cost of the system. Based on the redundant degree-of-freedom manipulator optimization strategy, it can be adjusted and reorganized as needed, saving workspace space and conducive to space optimization in different working environments. Each module is independent of each other to improve the reliability of system operation, and can also be upgraded and expanded separately according to standardized interfaces to meet the needs of technology development and changing needs.

1.3.2 Research Methodology

(1) Structural Design

According to the mortise and tenon system in traditional Chinese wooden buildings, the modular joint mechanical structure of each joint was designed, and the 3D solid model was established with SolidWorks software to select the driving device that met the task requirements. The joint modules were connected by steel standard parts, and different end-effectors with modular standard interfaces were designed according to the corresponding task scenarios In the servo control system, Raspberry Pi Pico W SMT package development board based on RP2040 chip is selected as the control carrier, the required component model and package are selected, the signal transmission protocol and interface are determined, and the PCB board supporting multi-channel goldfinger interconnect is designed by EDA software Based on the MicroPython development tool, a software platform was built to integrate all parts of the simple open-loop control of the robot arm test prototype, and complete the overall assembly of the modular series robot arm in multiple scenes

(2) Analysis and Optimization

Simulation plug-in in SolidWorks software was used to carry out static finite element analysis of the main stress components and output the data cloud map, evaluate the mechanical properties of 3D printing modular components made of photosensitive resin materials for testing the prototype, list and analyze the characteristics of modular assembly of the robot arm, and find out the general formula representing the total number of assemblies The kinematics equation of the robot arm was obtained based on the coordinate transformation method, and the entity model of the robot arm was built with Robotics Toolbox for MATLAB. The forward and inverse kinematics solution for the specific modular assembly form of the test prototype was completed through fifth-degree polynomial fitting, and the trajectory equation of the robot arm was optimized The energy consumption and space optimization of the redundant degrees of freedom of the robot arm under different assembly modes were established, and the multi-angle analysis and evaluation of the robot arm was achieved in the pre-processing stage, which provided a train of thought for the optimal solution of modular recombination under different scenarios.

(3) Functional Verification

The main mechanical structure of the manipulator (modular components of each joint) was 3D printed with light curing technology, and the drive device and PCB circuit board were assembled to verify the feasibility of the multi-channel modular assembly strategy. The kinematics solution algorithm was transferred to the software debugging platform, and the spatial pose and running trajectory of the end actuator of the manipulator were simply controlled The redundancy degree of freedom optimization method of the preliminary study was used to pretreat the assembly structure of the robot arm for specific task scenarios, and the feasibility of optimizing the application range and work efficiency of the robot arm through modular recombination under different scenarios was verified

1.3.3 Summary of the Author's Learning and Research Experience

The subject was completed independently by the author after self-study of relevant extra-curricular knowledge in high school, and the report inevitably contains errors and omissions of knowledge. Please kindly understand that the basic process of self-study and bibliography, as well as the achievements of research in related professional fields are attached at the end of the report, as evidence to prove the authenticity of the independent completion of the subject.

2. Multi-scene Modular Series Manipulator Structure Design

2.1 Working Principle and Layout Design

2.1.1 Analysis of Working Principle

The traditional discrete rigid tandem robotic arm is a bionic structure designed based on the human upper limb. It is composed of the basic mechanism, which usually includes seven parts: the base joint connecting rod, the arm end effector drive device and the control system.In order to achieve modular assembly and flexibly adjust the degree of freedom of each joint, the main connection components of each module are designed with the connecting rod that connects the joint to make it move relative as the starting point, and the installation position of the driving device is reserved in the movable joint part. The interface is unified in the base and connecting rod, and the connecting rod and the end actuator part, so as to realize the function of flexible connection of each module.

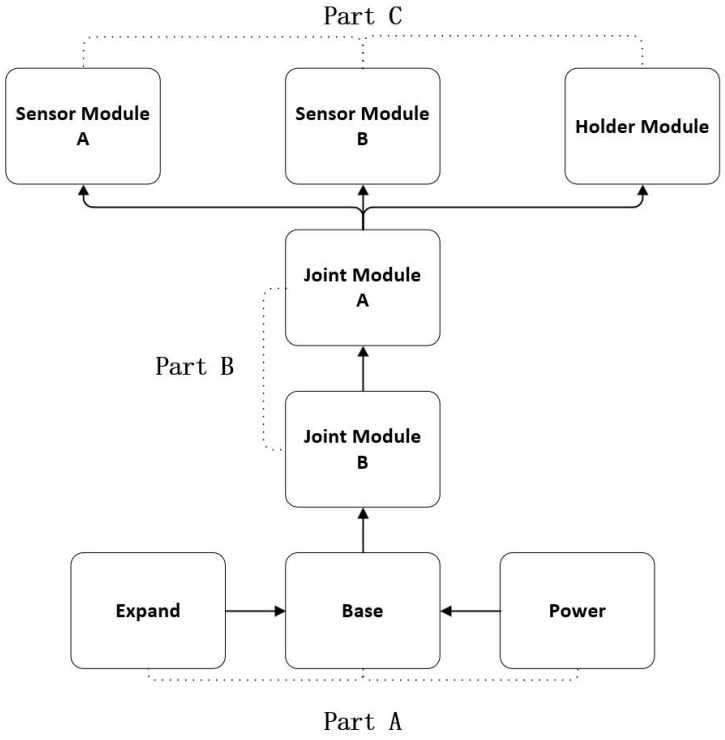


Figure 2-1 Modular layout design

2.1.2 Modular Layout Design

According to the modular connection idea mentioned in the previous section, the whole robot arm is divided into three parts: the end execution module of the base joint module. The base part is reserved for expansion interface, which can connect to the embedded controller to expand the optimization function of the robot arm such as deep learning. Power supply ports are reserved to ensure the stable operation of the drive device and control system when the load is large. The joint module consists of several sub-modules. Each sub-module can be connected in any direction and in any sequence according to different connection standards of the drive device The end execution module consists of a mechanical fixture and a sensor module with several kinds of sensors as the core, one of which is selected according to the needs of the scene during the working process and connected

2.2 Modular Connection Principle and Design

2.2.1 Modular Connection Principle

The modular connection component must ensure that the installation and disassembly process is simple and fast, and the connection is stable during use, which is reminiscent of the mortise and tenon joints in traditional Chinese ancient buildings Mortise and tenon joints are an important part of wooden structures in ancient buildings. Due to their unique connection mode and unique mechanical properties of wood, they can not only withstand certain bending moments, but also produce certain deformation, with typical semi-rigid connection characteristics [5]2

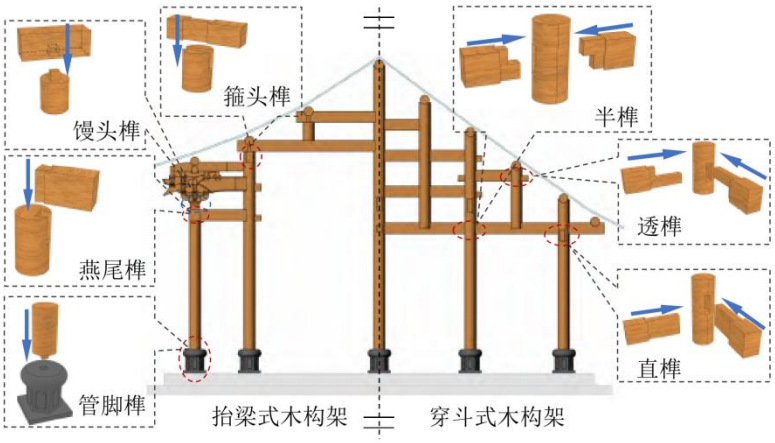
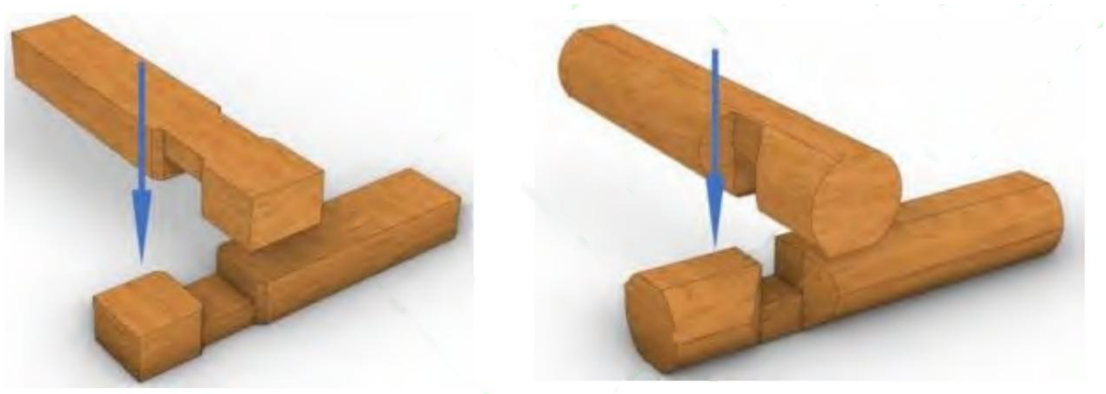


Figure 2-2 Schematic diagram of mortise and tenon joints in a timber frame[5]2

As shown in Figure 2-2, the dovetail joint is usually put and put, that is, the tenon part is made wide at the head and narrow at the root. Therefore, the dovetail joint has a good tensile strength and is widely used in the parts that need to be tensioned, as shown in Figure 2-3, while the cross cut half tenon is used for the lap joint between square horizontal members, and its function is to connect two pieces of wood frame, and enhance the stability and lateral resistance of the wooden frame. [5]3



(a) Cross cut half tenon (b) Cross waist tenon

Figure 2-3 Mortise and tenon joints connecting horizontal members[5]3

According to the requirements of modular connection of mechanical arm, combined with the characteristics of wide dovetail tenon and narrow root and cross cut half tenon used for lap between square horizontal members, T and N connection nodes are designed, wherein T connection node is located at the top of the connecting component, and N connection node is located at the bottom of the connecting component.

2.2.2 Modular Connection Design

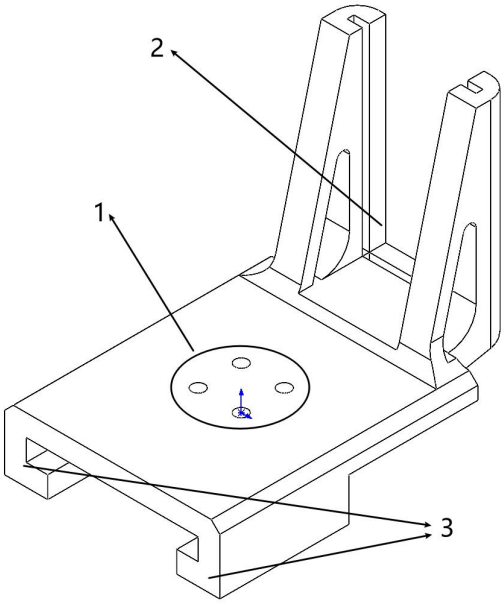


Figure 2-4 Modular connecting assembly bottom

1：M3　2：PCB groove　3：N connection node

As shown in Figure 2-4, there are four M3 screw holes with a diameter of 3mm and a depth of 3mm at the bottom of the modular connection assembly of the mechanical arm. The rear edge of the standard assembly of the drive device structure is designed with a PCB fixing slot with a width of 30mm and a height of 58.72mm, which is used to fix the sliding PCB board in the vertical direction.

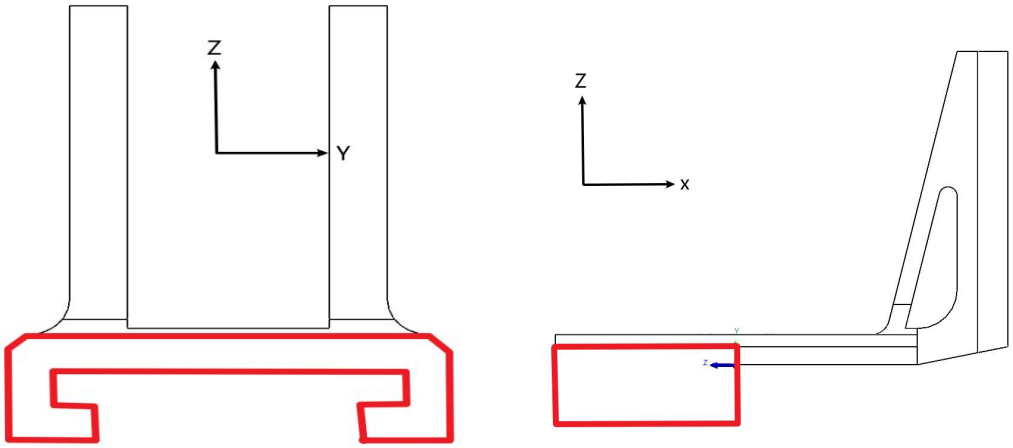


Figure 2-5 N connection node

As shown in Figure 2-5, the bold part is the main structure of N-shaped connection nodes, which can be mathematically abstracted as a fixed connection channel X_N. A coordinate system $\{N_i\}$ is established at the center of the entity, and the center of the standardized screw hole is defined as the origin O_{Ni}. The direction perpendicular to the fixed plane of the connecting component drive device is the Z axis, the direction parallel to the connection section is the X axis, and the direction perpendicular to the connection section is the Y axis The connecting node can move along the X axis to connect the T-shaped node.

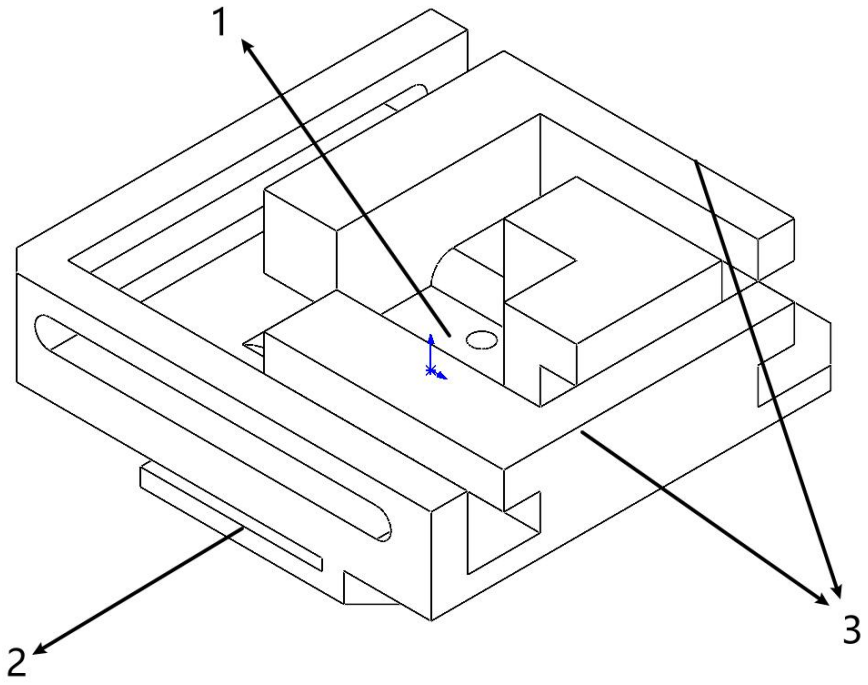


Figure 2-6 Modular connecting components top

1：M3　2：PCB groove　3：T connection node

As shown in Figure 2-6, the top of the modular connection assembly of the mechanical arm is designed with four M3 screw holes with a diameter of 3mm and a depth of 3mm, which are used to fix the drive device structure. The rear edge of the standard assembly is designed with an L-shaped PCB fixing slot for fixing the sliding PCB board in the horizontal direction.

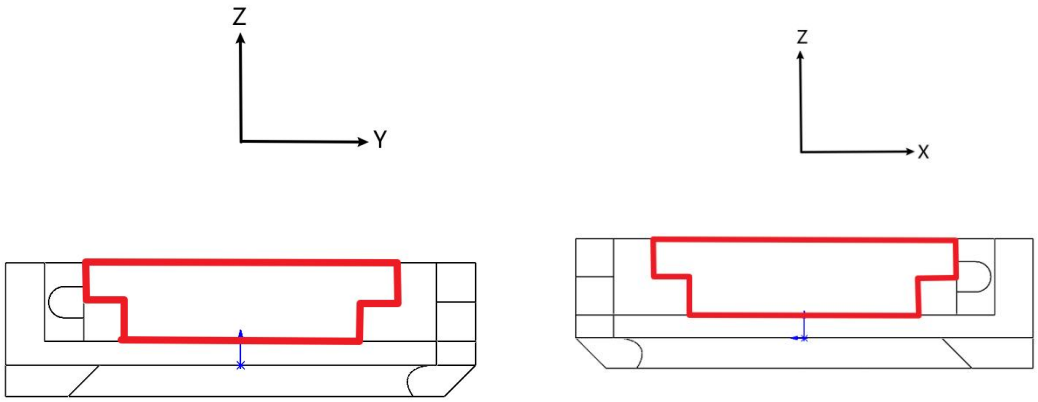


Figure 2-7 T connection node

The bold part is T, as shown in Figure 2-7 The main structure of the font connection node can be mathematically abstracted as an optional connection channel X_N and a coordinate system $\{N_i\}$ is established at the center of the entity, and the origin O_{Ni} of the center of the standardized screw hole is defined as the Z axis perpendicular to the fixed plane of the connecting component drive device, the X axis parallel to the front view connection channel section, and the Y axis perpendicular to the front view connection channel section.

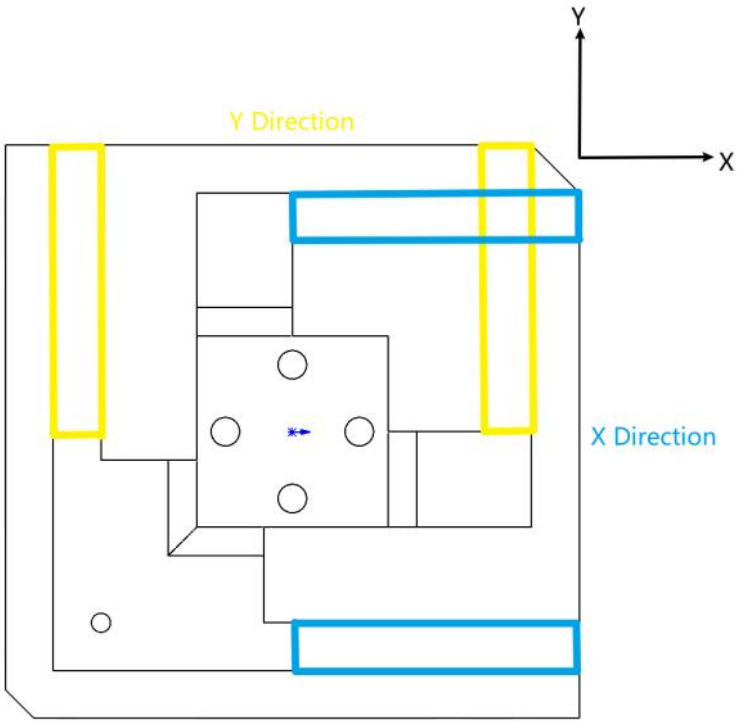


Figure 2-8 Top view of the T node

As shown in Figure 2-9, the N-shaped connection node can move along the connection channel of the T-shaped connection node system in the X direction and the Y direction, and the two connection nodes can be flexibly connected in two directions perpendicular to each other, thus achieving the purpose of increasing the degree of freedom of the manipulator arm by means of modular reorganization.

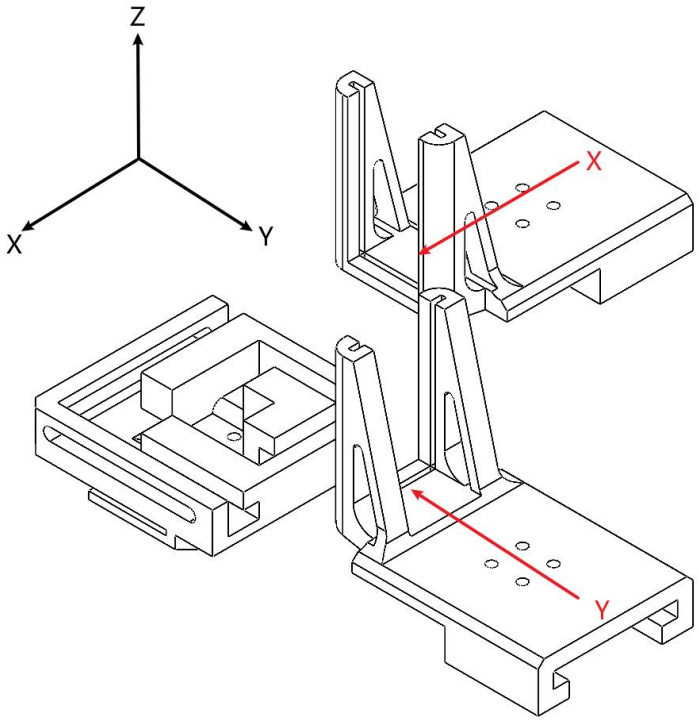


Figure 2-9 Schematic diagram of the connection between the T and N connection nodes

2.3 Modular Connection Principle and Design

2.3.1 Selection and Modeling of the Driving Device

To ensure the smooth operation of each joint of the mechanical arm, taking into account the cost of the prototype, the driving device selected 15kg.cm torque PWM signal digital steering gear of Zolling Technology, as shown in Figure 2-13.

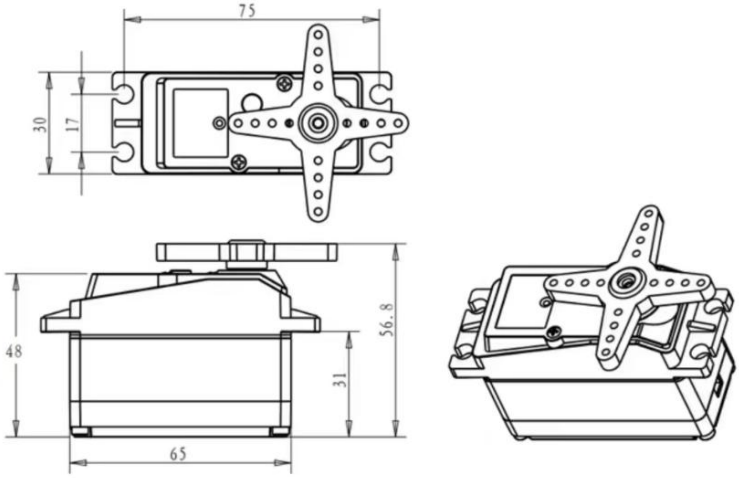


Figure 2-10 15kg.cm torque, PWM signal, digital servo unit: mm

Quality	Parameter
Turning Speed	0.16sec/60°(6V)
Servo Accuracy	0.24°
Stall Torque	15kg.cm
Number of Gear Teeth	25T
Gear Type	Metal
Supply Voltage	5-9V
Angle of Rotation	180°
Control Mode	PWM Pulse Signal
Pulse Wdth Range	0.5ms-2.5ms
Control Cycles	20ms
No-load current	100mA
Product line length	30cm

Table 2-11 Basic parameters of the servo

Operating Voltage	Standby Current	No-load Speed
6V	4mA	0.17sec/60°
7.4V	5mA	0.15sec/60°
8.4V	6mA	0.13sec/60°

Table 2-12 Servo current parameters

Assemble servo 90° and 45° biaxial brackets, steel standard parts and build a 3D solid.

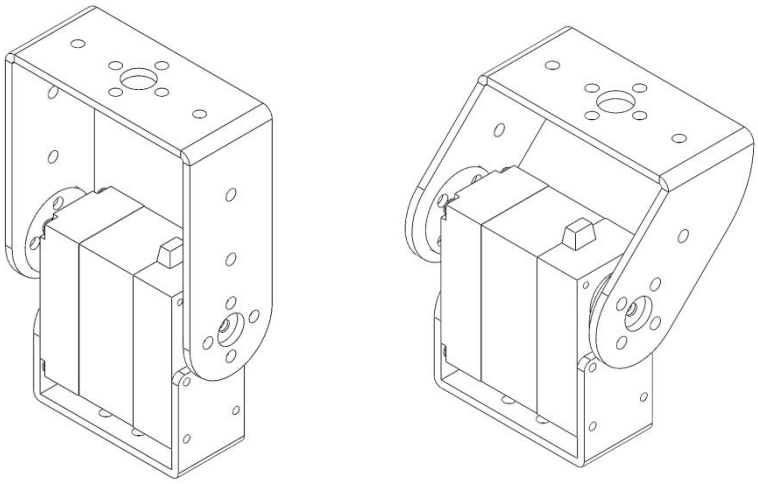


Figure 2-13 3D model of the standard parts of the servo assembly

2.3.2 Design of Control System

(1) Selection of Controllers

The servo control system selects the Raspberry Pi Pico W SMT package development board based on the RP2040 chip as the control carrier, equipped with dual-core ARM Cortex M0 + processor, running frequency up to 133MHz flexible clock, built-in 264KB SRAM and 2MB on-chip Flash, equipped with 2 SPI,2 I2C,2 UART,3 12-bit ADCs,16 controllable PWM channels, accurate on-chip clock and timer temperature sensor Onboard 2.4GHz wireless interface chip (802.11n or Wifi4), can carry out TCPIP protocol network communication, expand the network control related functions

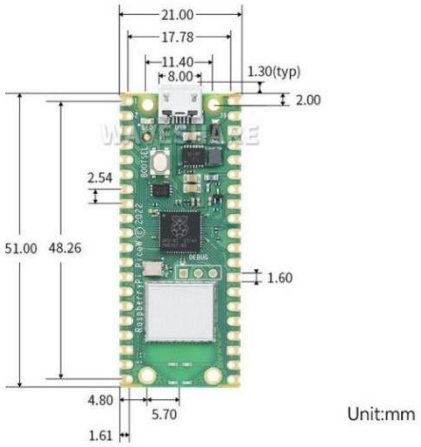


Figure 2-14 Raspberry Pi Pico

(2) Motherboard interface definition

In addition to the basic signal interface on the Base motherboard equipped with the control carrier, an SPI protocol interface and an I2C protocol interface are also extended to connect the embedded controller to expand the deep reinforcement learning function of the robot arm. The power supply port is reserved for external power supply to ensure the stable operation of the drive device and control system when the load is large.

(3) Definition of the Interface of Each Module

The Joint A and Joint B connecting boards for each joint module are equipped with three PWM digital interfaces, five LED status indicators and three steering gear signal selectors respectively, which are used to transmit stable PWM signal for the steering gear to feedback the real-time status of the manipulator arm and select the signal interface of the steering gear.

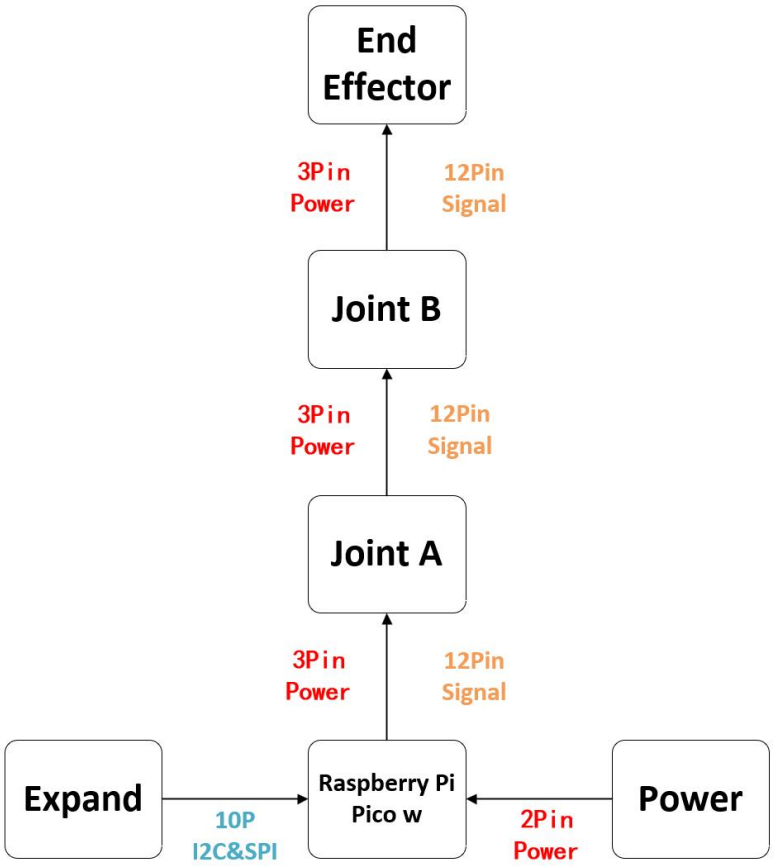


Figure 2-15 Structure diagram of PCB signal and power interface

The HolderSensor connecting board connected to the end actuator is equipped with a 2-way ADCDC analog-to-digital conversion interface for connecting the analog sensor and a 3-way digital PWM interface for connecting the digital sensor and the mechanical fixture driven by the steering engine.

(4) Selection of Component Packaging

The 16P 2.54mm package Goldfinger plug-in and socket are used to connect each module, and the 12P 1mm package FPC cable and XH 3P 2.54mm package terminal cable are used to connect each module, ensuring the stability of power supply and signal transmission and the convenience of disassembly and installation to determine the function of each connection board After the packaging of each component and the power supply and signal interface, the PCB board drawing is drawn.

2.3.3 Design of End Effectors

The end effector needs to complete specific work in different scenarios to ensure the spatial adaptability of the robot arm in different environments. The modular design enables the robot arm to quickly switch between different tasks, thereby improving production efficiency and flexibility. Each scene module is fixed on the same modular component as the joint module, and the interface is unified to achieve the purpose of flexible combination of various parts.

(1) Parallel Fixture Module

As shown in Figure 2-16, the parallel fixture is used for grasping rigid and flexible objects with a diameter of 50mm, and can complete specific tasks in automatic manufacturing, rescue, medical and other scenarios.

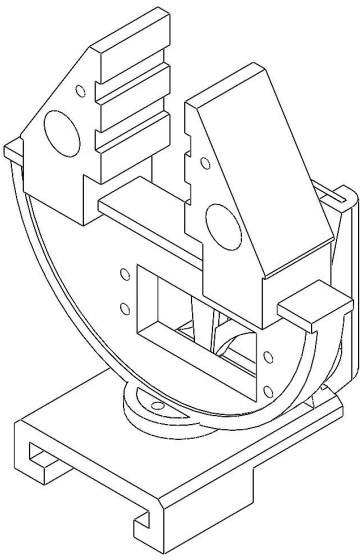


Figure 2-16 Parallel fixture module

(2) Ultrasonic Detection Module

As shown in Figure 2-17, the module is equipped with a digital ultrasonic sensor of HC-HR04 model, which can detect the current distance of the environment where the end actuator of the robot arm is located and output accurate data according to the digital signal to complete specific tasks such as medical rescue and special environment detection.

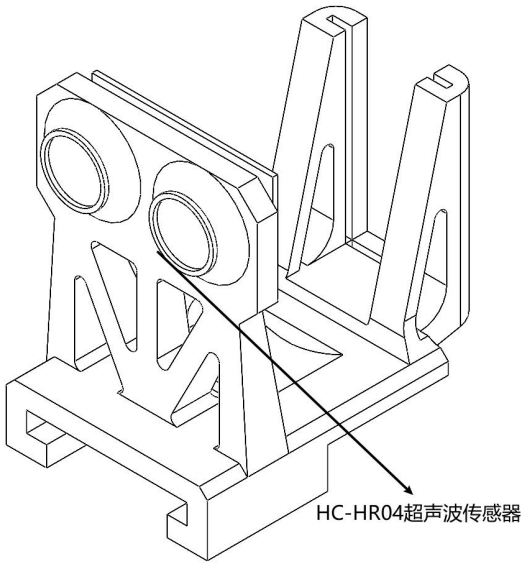


Figure 2-17 Ultrasonic detection module

(3) Gas Composition Detection Module

As shown in Figure 2-18, the module is equipped with an analog gas component sensor of the MQ series, which can detect the concentration of specific gas components in the environment of the end actuator of the robot arm and output accurate data according to the analog signal, and can complete specific tasks in rescue, automatic manufacturing, special environment detection and other scenarios.

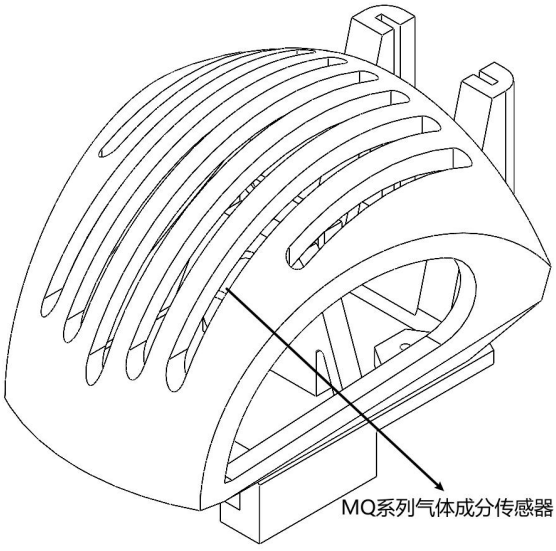
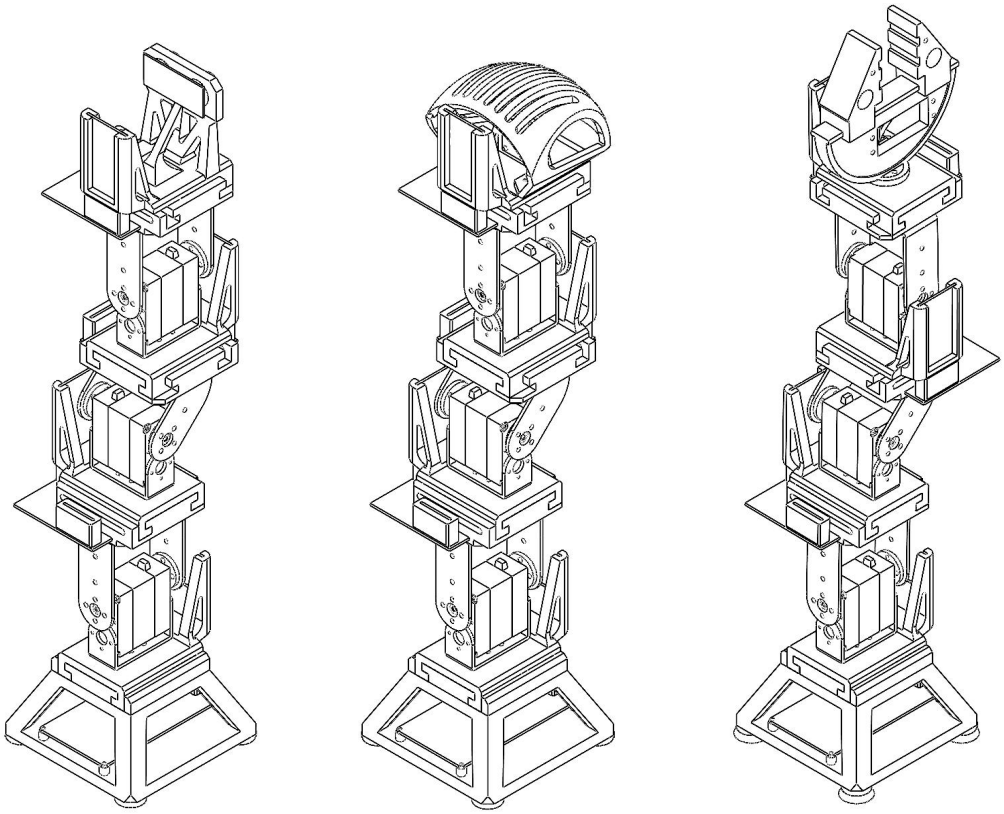


Figure 2-18 Gas composition detection module

2.3.4 Overall Structural Assembly

In the test, 1 45-joint module, 2 90-joint modules and base module were selected to assemble the fixture module, ultrasonic detection module and gas component detection module. The whole system could be stably connected and easily assembled and reassembled, achieving the purpose of building the basic structure of multi-scene modular series manipulator.



(b) Parallel fixture module *(c) Ultrasonic Detection Module* *(a) Gas composition detection module*

Figure 2-19 Diagram of the total assembly

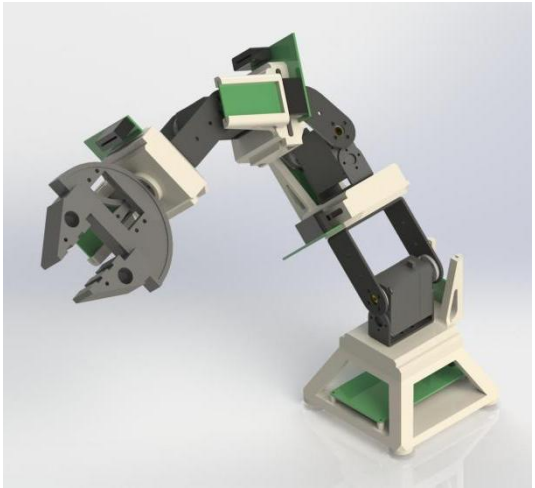


Figure 2-20 General assembly rendering

3. Analysis and Optimization of Multi-scenario Modular Tandem Manipulators

3.1 Finite Element Analysis of Static Stress of Manipulator

3.1.1 Selection and Modeling of the Drive Device

Initially, finite element method was limited to solving some mechanical problems in the field of structure. However, with the continuous evolution of this technology, finite element method has been widely used in many fields, including fluid mechanics, electromagnetic field and thermodynamics When dealing with structural problems, finite element method usually involves the following steps [6].

(1) Structural Discretization

Firstly, the structure is divided into several units. In order to connect these units together, nodes need to be introduced. By discretizing the structure, multiple units subjected to load at the joint of nodes can be obtained, and these units eventually form a whole structure.

(2) Unit Analysis

The element stiffness equation is formed by combining physical and geometric relations through the virtual work principle. In the initial case, there is no force on the node. In order to introduce the node force, all the loads acting on the element can be transferred to the node equivalent.

(3) Overall Analysis

When analyzing the force on each node of the element, it is also necessary to consider the load outside the node and the force exerted by each element on the node. Finally, these factors can be used to derive the overall stiffness equation through the

node displacement as a variable.

(4) Constraints Are Introduced to Calculate the Displacement Solution

In the equation of the overall stiffness after the above steps are determined, a support condition is added to solve the equilibrium equation and displacement.

(5) Calculate Other Quantities

Other quantities such as stress and strain can be calculated from calculated node displacements and combined with physical and geometric relationships.

3.1.2 Static Analysis of Each Joint and Overall of the Robotic Arm

Static analysis is the basis of finite element analysis Because the joint module of the manipulator is affected by load or operating environment during operation, which may lead to wear or fracture of the big arm, it is particularly important to carry out static analysis of each joint module and the whole to ensure that it is kept within the safe range during the whole operation [7].

(1) Basic Steps

(a) Establish a Finite Element Model with a Reasonable and Simplified Structure

(b) Define Material Properties

The material for statics analysis is defined as 9400A photosensitive resin, and the basic parameters are shown in Table 3-1.

Quality	Value	Unit
Elastic Modulus	2750000000	N/m^2
Poisson's Ratio	0.44	
Mass Density	1150	kg/m^3
Tensile Strength	28000000	N/m^2
Compressive Strength	70	N/m^2
Thermal Conductivity	0.188	W/(m·K)

Table 3-1 Basic parameters of 9400A

(c) Meshing

(d) Add Constraints and Loads

(e) Computational Solving

(2) Static Analysis of the Joint Base

(a) Meshing the Model

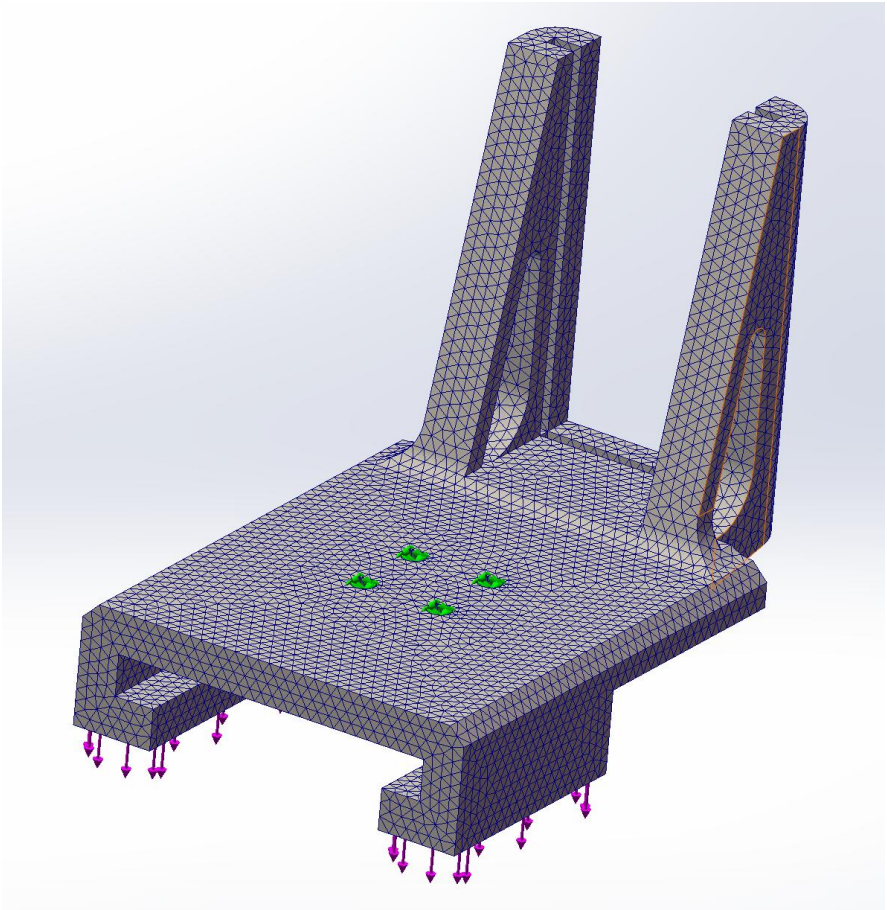


Figure 3-2 Finite Element Analysis Meshing at the Bottom of the Joint

(b) Fix 4 M3 Screw Holes, and Add a Force of 10N Vertically Downward on the Bottom Surface of the "N" Modular Connection Node

(c) Output of Static Stress Analysis Displacement Results

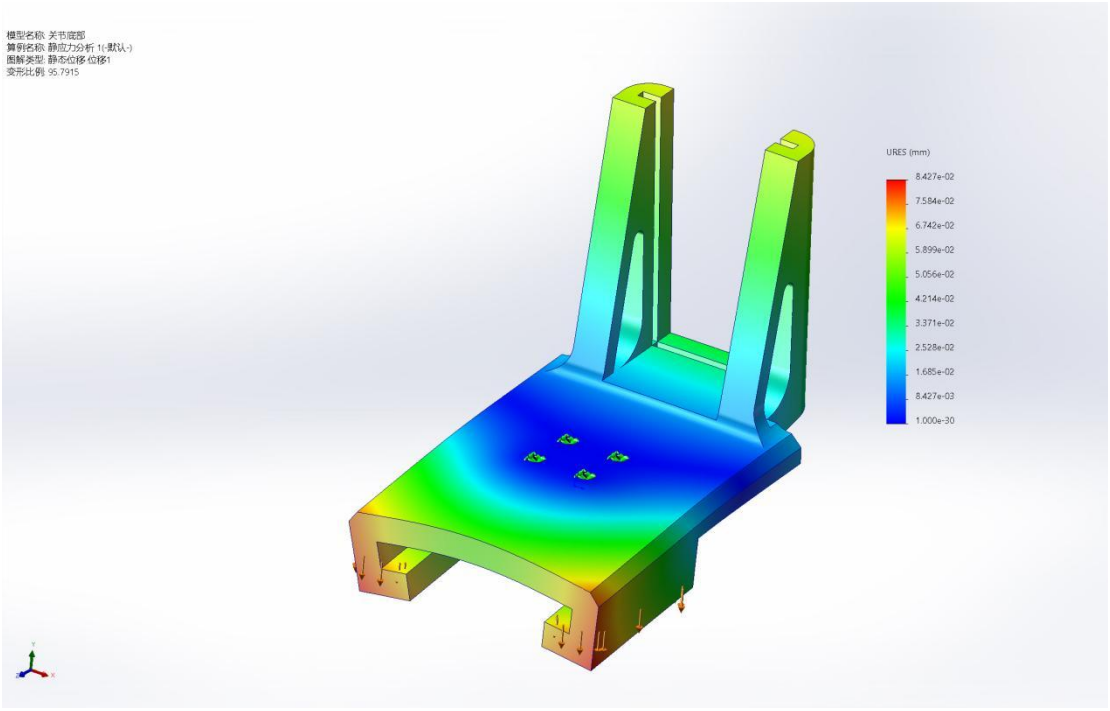


Figure 3-3 Static stress analysis of joint bottom displacement

(d) Output Static Stress Analysis Strain Results

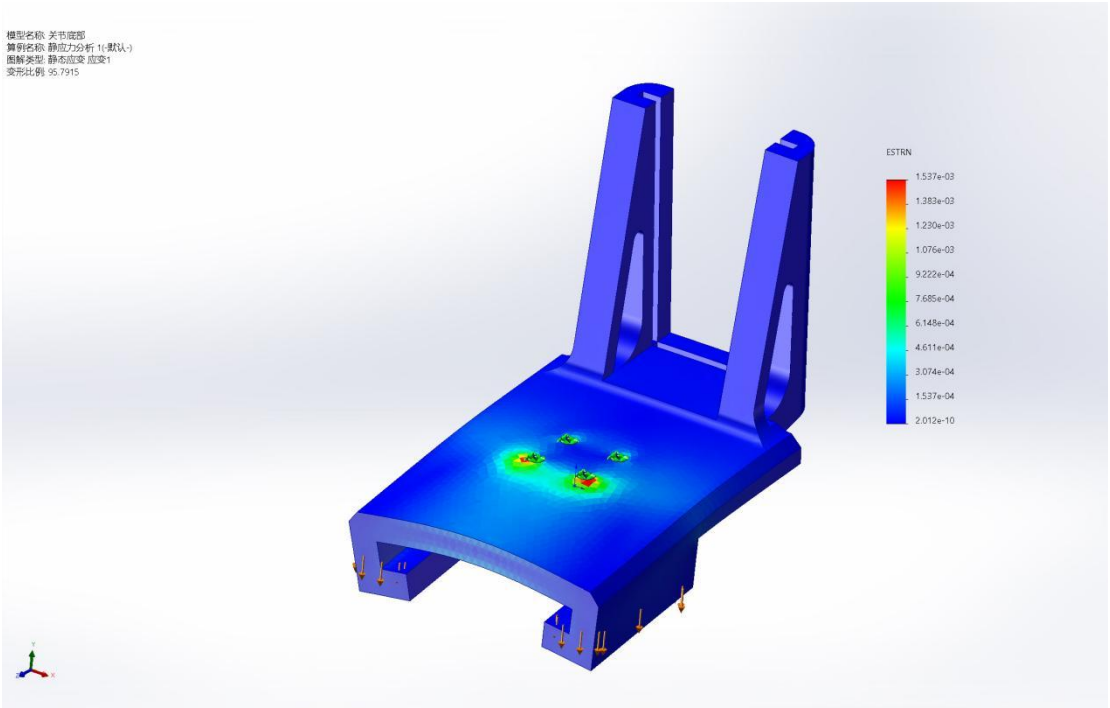


Figure 3-4 Strain results of static stress analysis of joint bottom

(e) Output of Static Stress Analysis Stress Results

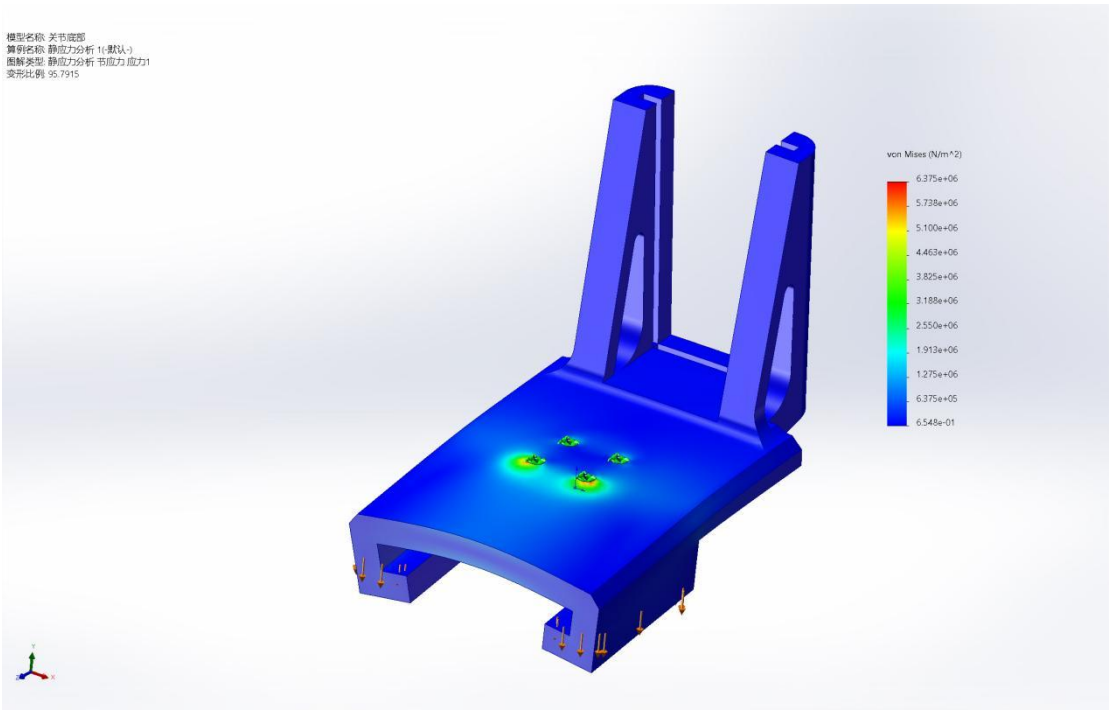


Figure 3-5 Static stress analysis stress results at the bottom of the joint

(3) Static Analysis of the Top of the Joint

(a) Meshing the Model

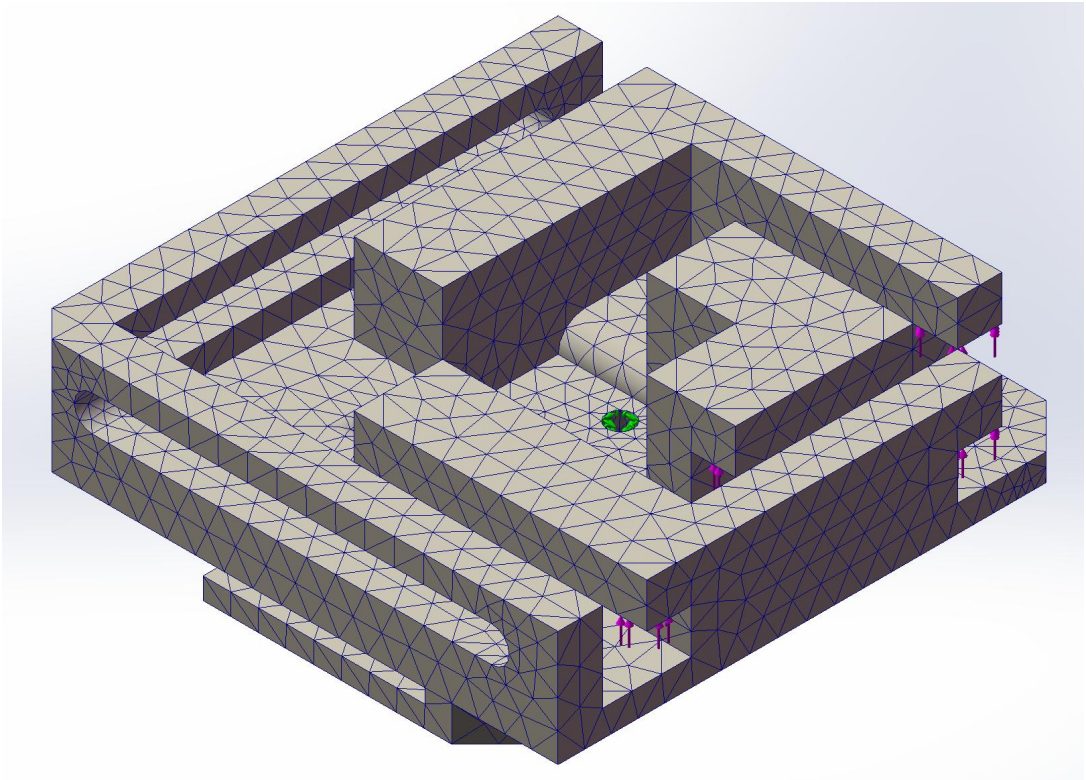


Figure 3-6 Joint top finite element analysis meshing

(b) Fix 4 M3 Screw Holes and Add a Force of 10N Vertically Upwards to the Connecting Surface of the T Modular Connection Node.

(c) Output of Static Stress Analysis Displacement Results

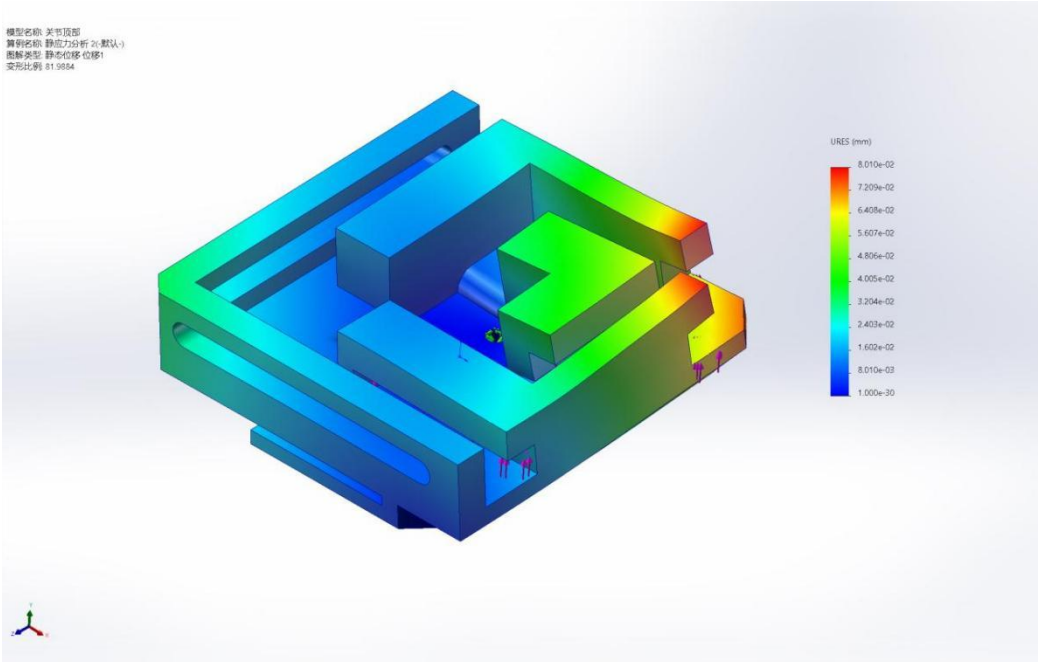


Figure 3-7 Static stress analysis of joint bottom displacement results

(d) Output Static Stress Analysis Strain Results

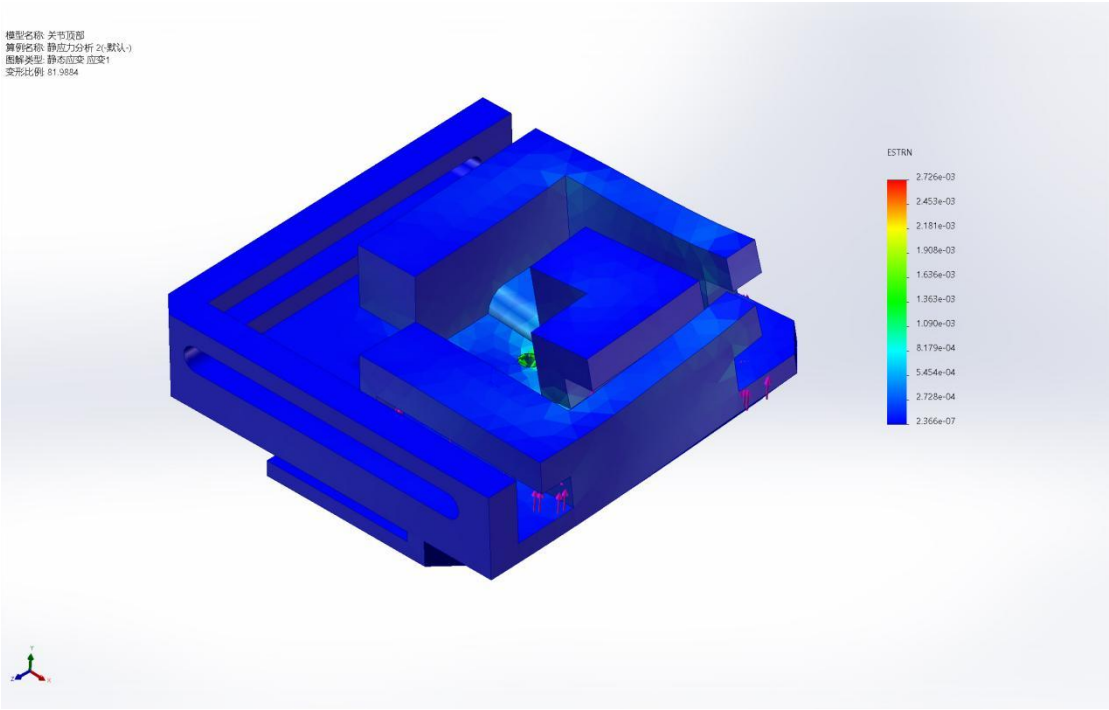


Figure 3-8 Strain results of static stress analysis at the top of joint

(e) Output of Static Stress Analysis Stress Results

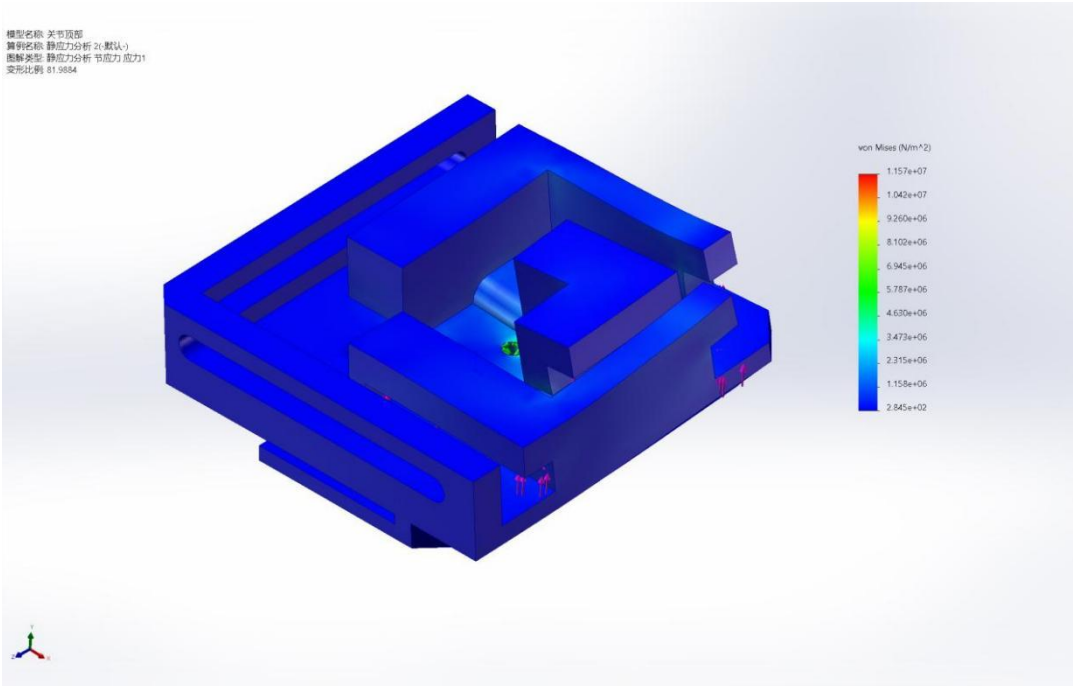


Figure 3-9 Static stress analysis of joint top stress results

(4) Static Analysis of the Top Surface of Each Joint as a Whole

(a) Model Mesh Division

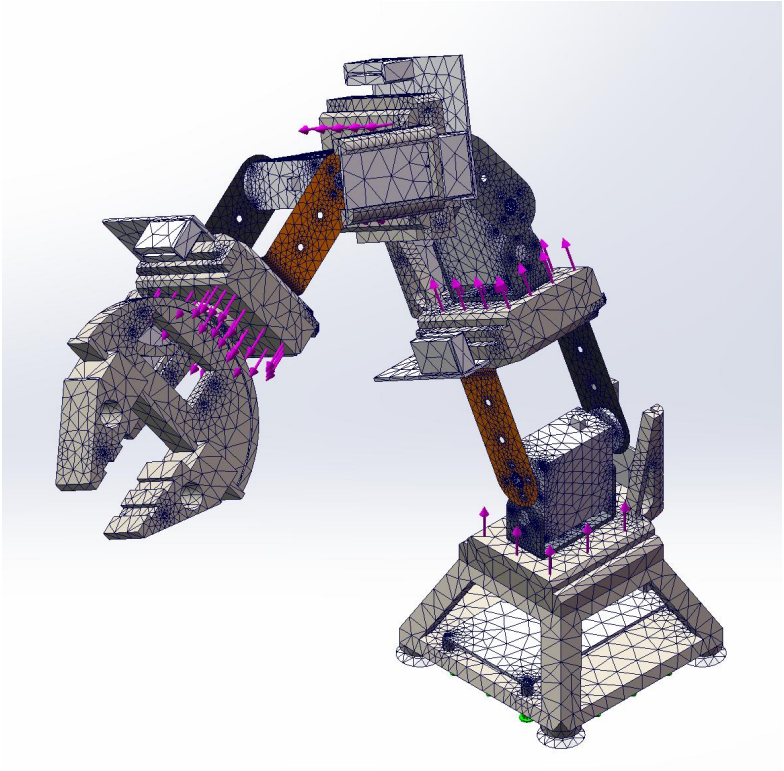


Figure 3-10 Meshing of mechanical arm by finite element analysis

(b) Fix the Base Ground and a Force of 10N Vertically Upwards on the Top Surface of Each Module

(c) Output of Static Stress Analysis Displacement Results

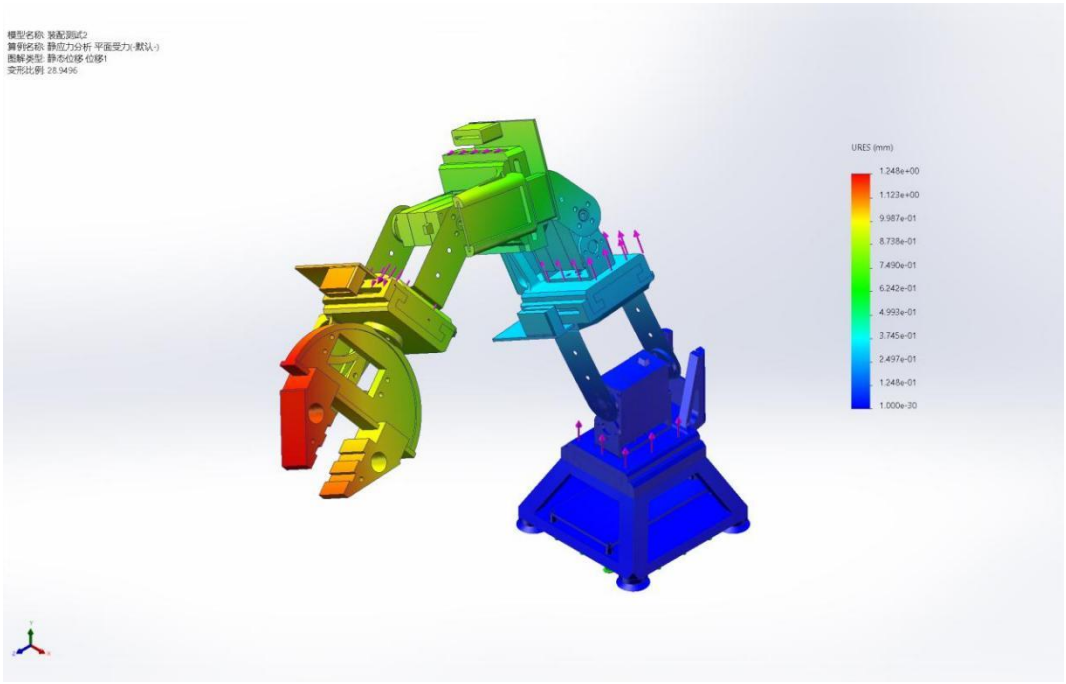


Figure 3-11 The displacement results of the whole static stress analysis of the manipulator

(d) Output Static Stress Analysis Strain Results

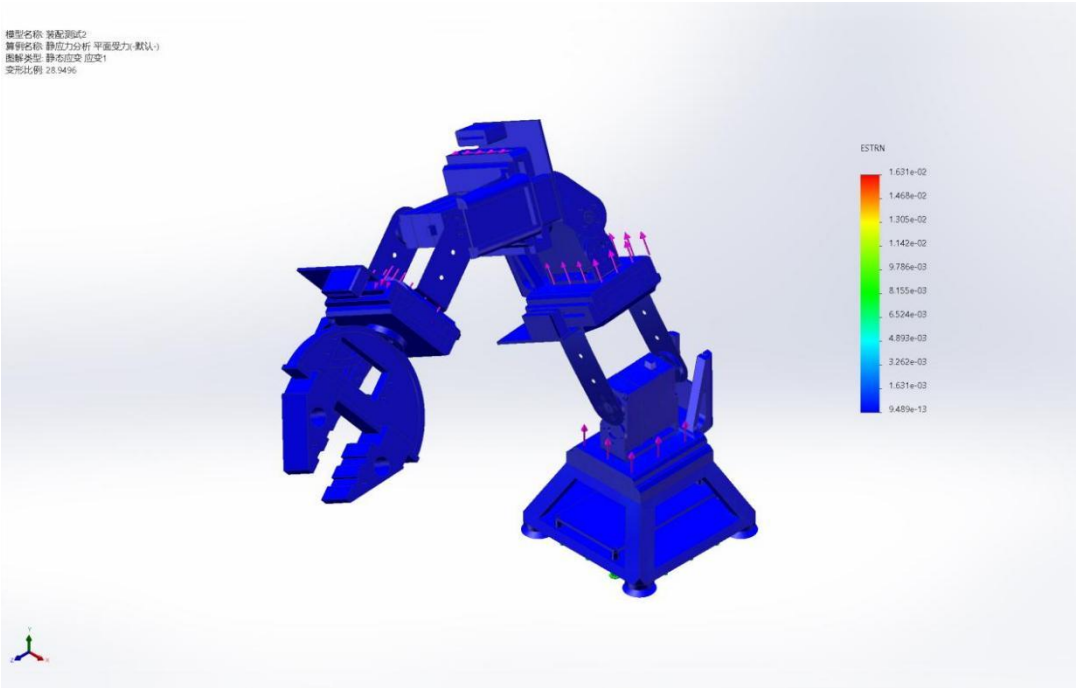


Figure 3-12 Strain results of static stress analysis of the whole mechanical arm

(e) Output Static Stress Analysis Stress Results

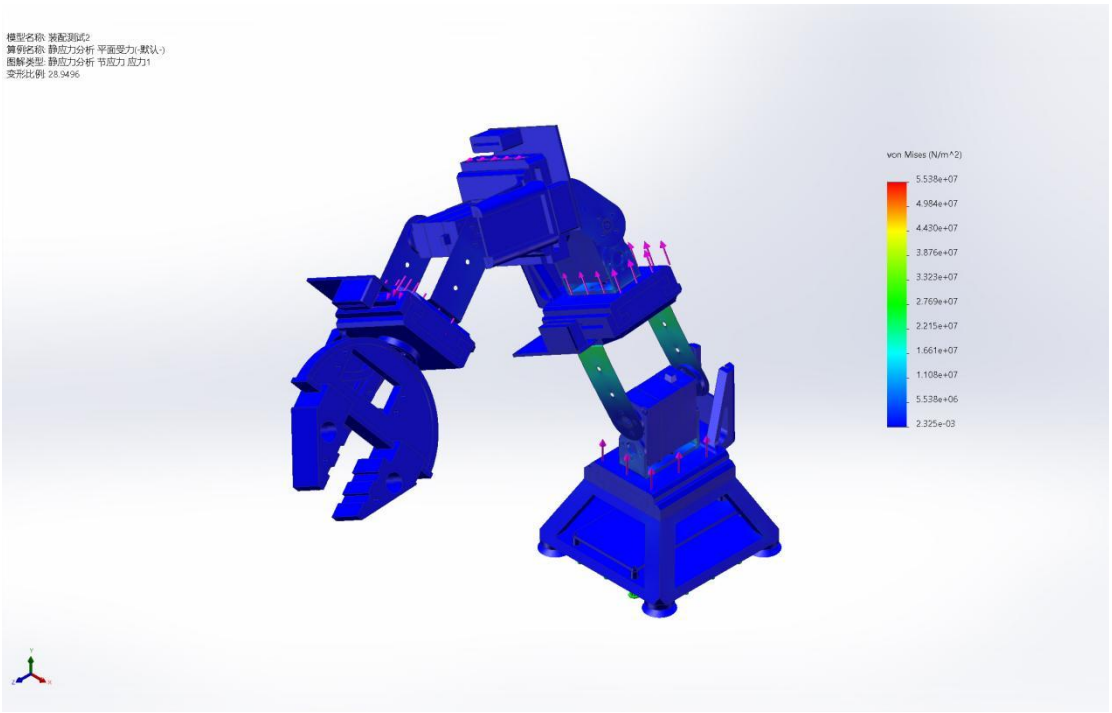


Figure 3-13 The result of static stress analysis of the whole mechanical arm

3.1.3 Conclusion of Static Analysis

When 10N load is applied to the top surface of each joint of the multi-scene modular series robot arm, it can work normally, and the displacement and deformation are within the tolerable range of the robot arm.

3.2 Analysis of Modular Assembly of Robotic Arms

3.2.1 Overview of Modular Assembly Analysis

The arm prototype used for testing consisted of a base, a 45-joint module and two 90-joint modules. The end-effector related modules were omitted to simplify the process.

3.2.2 Establish a World Coordinate System

The pose of the first joint module connected to the known base is fixed. The world coordinate system {O} is established by taking the line where the joint axis of the module is located as the Y axis, the line perpendicular to the joint axis is the X axis, and the line perpendicular to the top plane is the Z axis.

3.2.3 Define the Pose of Each Module

Two 90-joint modules and one 45-joint module are defined as:

$$A\ B\ C$$

Joint modules with joint axes perpendicular to the X-axis are denoted as:

$$X_y$$

Similarly, from bottom to top, a joint module with a joint axis perpendicular to the X axis, a joint module B with a joint axis perpendicular to the Y axis, and a C joint module with a joint axis perpendicular to the X axis are denoted as:

$$A_yB_xC_y$$

3.2.4 Calculate the Total Number of Analysis Assembly Types

Known The first joint module of the known base connection is fixed, and the remaining two joint poses can be arbitrarily changed to obtain the total number of assembly types.

$$N_3 = C_3^3 \cdot 3! \cdot 2^{3-1} = 24$$

(a) ABC:

$$A_yB_xC_x \quad A_yB_yC_y$$

$A_yB_xC_y$ $A_yB_yC_x$

Figure 3-14 shows the simplified model

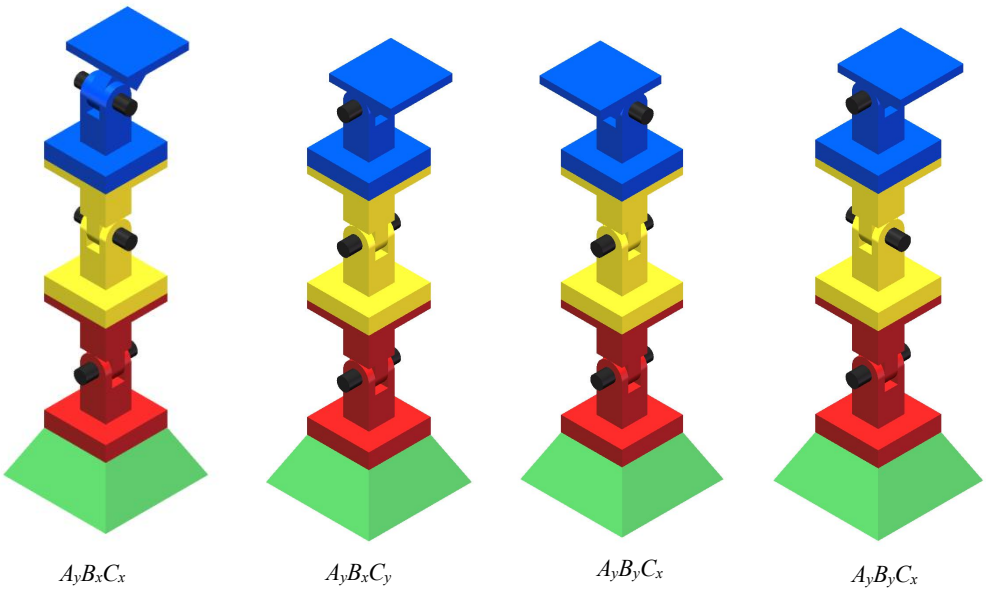


Figure 3-14

(b) ACB:

$A_yC_xB_x$ $A_yC_yB_y$

$A_yC_xB_y$ $A_yC_yB_x$

Figure 3-15 shows the simplified model

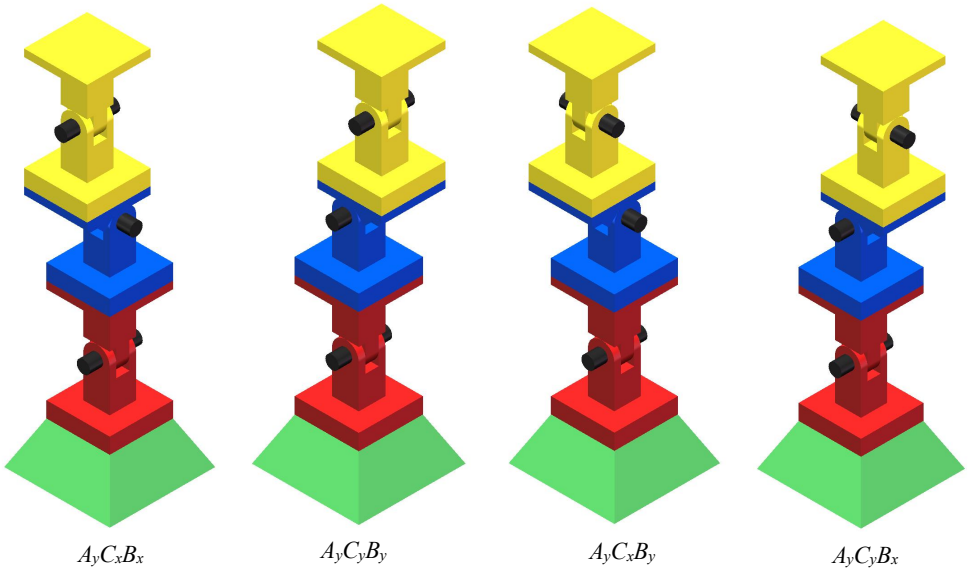


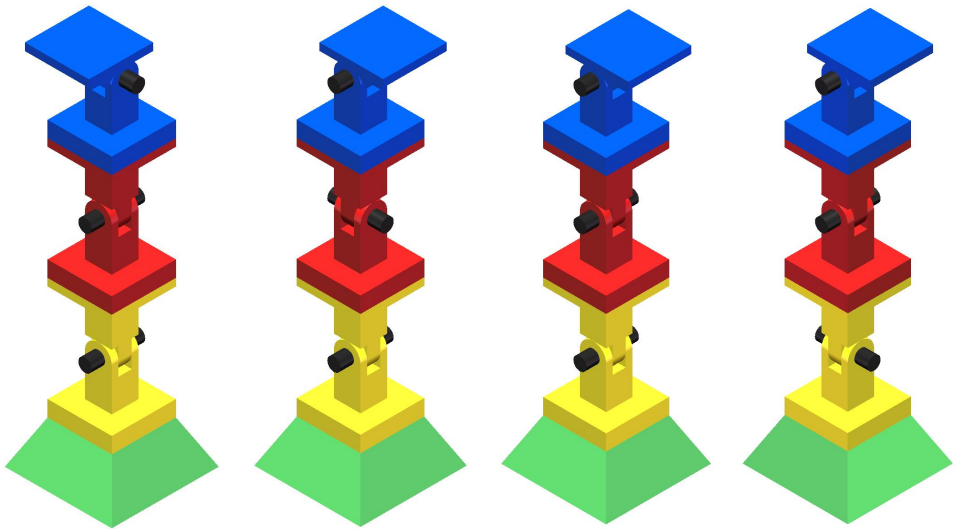
Figure 3-15

(c) BAC:

$B_yA_xC_x$ $B_yA_yC_y$

$B_yA_xC_y$ $B_yA_yC_x$

Figure 3-16 shows the simplified model



$B_yA_xC_x$ $B_yA_yC_y$ $B_yA_xC_y$ B_yA_yC

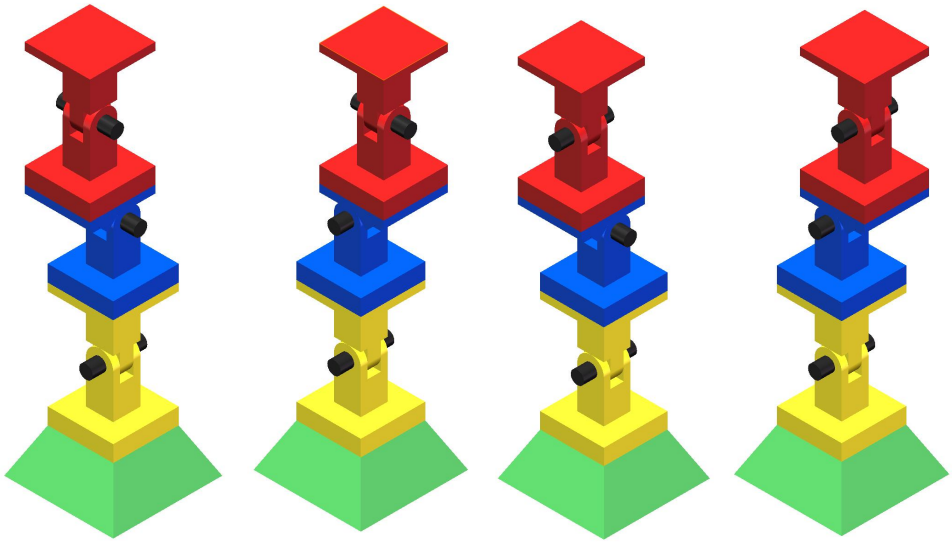
Figure 3-16

(d) BCA:

$B_yC_xA_x$ $B_yC_yA_y$

$B_yC_xA_y$ $B_yC_yA_x$

Figure 3-17 shows the simplified model



$B_yC_xA_x$ B_yC_yA $B_yC_xA_y$ $B_yC_yA_x$

Figure 3-17

(e) CAB:

$C_yA_xB_x$ $C_yA_yB_y$

$C_yA_xB_y$ $C_yA_yB_x$

Figure 3-18 shows the simplified model

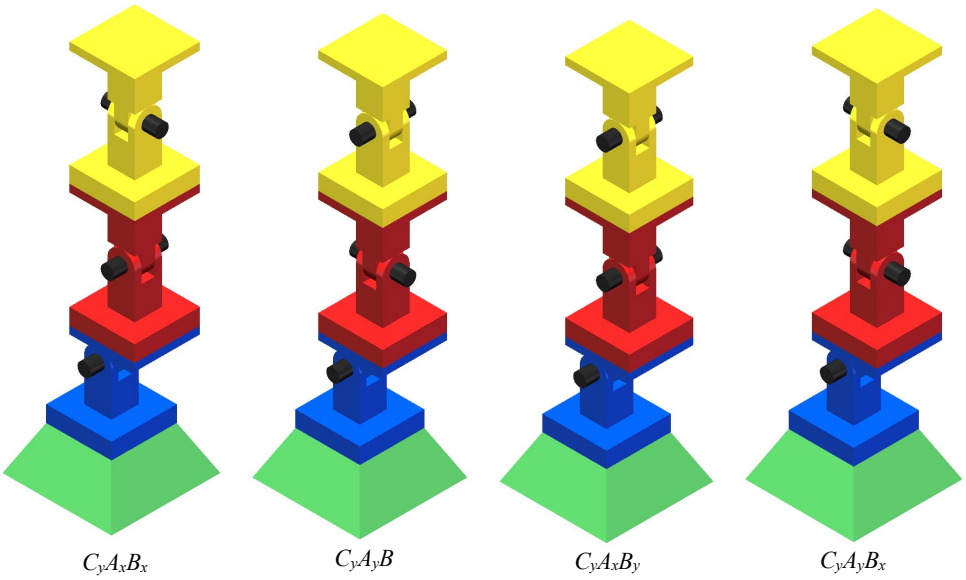


Figure 3-18

(f) CBA:

$C_yB_xA_x$ $C_yB_yA_y$

$C_yB_xA_y$ $C_yB_yA_x$

Figure 3-19 shows the simplified model

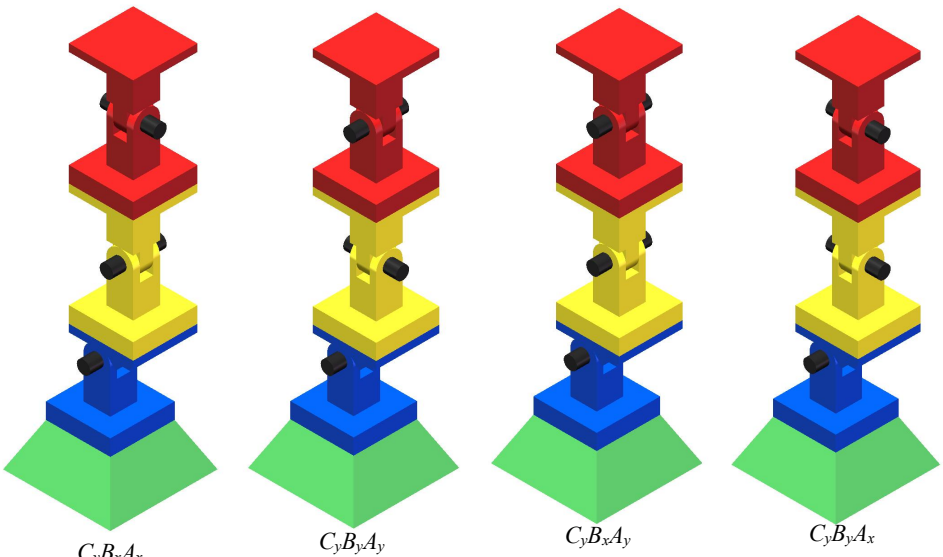


Figure 3-19

3.2.5 Conclusion of Assembly Analysis

To sum up, there are 24 joint combinations of the robot arm used for prototype testing, and each joint and tool module can be flexibly disassembled and maintained separately, which greatly reduces the maintenance cost of the system, saves the working space and can be adjusted and restructured according to needs, and is conducive to the space optimization of different working environments.

3.3 Kinematic Modeling of Robotic Arms

3.3.1 Establishment of Linkage Coordinate System

Taking AyCxBy as an example, the mechanical fixture module is connected, which is called AyCxBy-D, and the parameter table of SDH standard is listed first.

i	*θ*	*d*	*α*	*a*	*θ Range*
1	θ_1	95	90°	0	(0°,0°)
2	θ_2	0	90°	113.5	(0°,180°)
3	θ_3	21.34	90°	88.5	(0°,180°)
4	θ_4	0	0	152	(0°,180°)

Table 3-20 AyCxBy-D SDH table unit:mm

The connecting rod coordinate system of AyCxBy-D manipulator is established

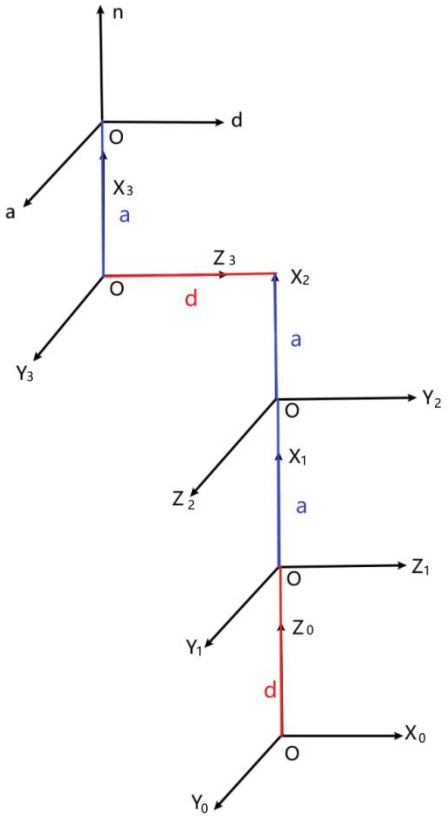


Table 3-21 AyCxBy-D manipulator linkage coordinate system

3.3.2 Kinematic Equations of the Robotic Arm

According to the parameters in Table 3-20, the transformation matrix of each connecting rod can be obtained as follows:

$$T_1^0 = \mathrm{Rot}(z,\theta_1)\mathrm{Rot}(x,\frac{\pi}{2})\mathrm{Trans}(0,0,d_1) \tag{3.2}$$

$$T_2^1 = \mathrm{Rot}(z,\theta_2)\mathrm{Rot}(x,\frac{\pi}{2})\mathrm{Trans}(a_2,0,0) \tag{3.3}$$

$$T_3^2 = \mathrm{Rot}(z,\theta_3)\mathrm{Rot}(x,\frac{\pi}{2})\mathrm{Trans}(a_3,0,0)\mathrm{Trans}(0,0,d_3) \tag{3.4}$$

$$T_4^3 = \mathrm{Rot}(z,\theta_4)\mathrm{Trans}(a_4,0,0) \tag{3.5}$$

Expand (3.2)、(3.3)、(3.4)、(3.5) to obtain:

$$T_1^0 = \begin{bmatrix} \cos\theta_1 & 0 & \sin\theta_1 & d_1\cdot\sin\theta_1 \\ \sin\theta_1 & 0 & \cos\theta_1 & -d_1\cdot\cos\theta_1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.6}$$

$$T_2^1 = \begin{bmatrix} \cos\theta_2 & 0 & 0 & a_2\cdot\cos\theta_2 \\ \sin\theta_2 & 0 & 0 & a_2\cdot\sin\theta_2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.7}$$

$$T_3^2 = \begin{bmatrix} \cos\theta_3 & 0 & \sin\theta_3 & a_3\cdot\cos\theta_3+d_3\cdot\sin\theta_3 \\ \sin\theta_3 & 0 & -\cos\theta_3 & -d_3\cdot\cos\theta_3+a_3\cdot\sin\theta_3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.8}$$

$$T_3^4 = \begin{bmatrix} \cos\theta_4 & -\sin\theta_4 & 0 & a_4\cdot\cos\theta_4 \\ \sin\theta_4 & \cos\theta_4 & 0 & a_4\cdot\sin\theta_4 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.9}$$

matrix of AyCxBy-D manipulator as follows:Multiply the change matrix of each connecting rod to obtain the transformation

$$T_4^0 = T_1^0(\theta_1) \cdot T_2^1(\theta_2) \cdot T_3^2(\theta_3) \cdot T_4^3(\theta_4) \tag{3.10}$$

3.3.3 Robotic Arm Modeling by MATLAB

According to the parameters in Table 3-20 and the linkage coordinate system in Figure 3-21, Robotics Toolbox for MATLAB was used to model the AyCxBy-D robotic arm, as shown in Figure 3-22.

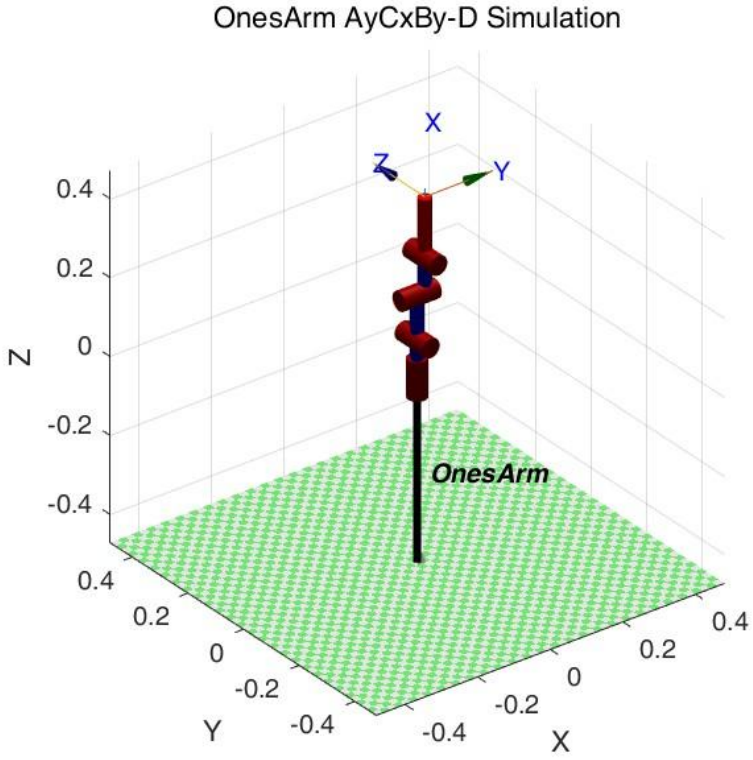


Figure 3-22 AyCxBy-D MATLAB model

3.4 Robotic Arm Redundant Degrees of Freedom

3.4.1 Analysis of Redundant Degrees of Freedom of Robotic Arm

A multi-scene modular series manipulator is a redundant manipulator. A redundant manipulator refers to a manipulator whose mechanical structure has more degrees of freedom than its motion space dimension. When its end can reach a large part of the working space, it is called a redundant manipulator.

Since the multi-redundancy robot arm has more degrees of freedom, the solution

of the inverse kinematics may produce multiple or even infinite solutions for the same end target in motion planning. For this reason, in addition to completing the end target task, the robot arm can also perform other additional tasks, such as avoiding obstacles, when realizing the motion planning of the robot arm Handle knot subtasks and pass through special waypoints [7].

In the process of multi-module assembly, the multi-scene modular tandem manipulator also has multiple solutions. This is because each pair of modular nodes has two connection channels perpendicular to each other (while the module connected to the base has only one connection channel), so the total number of assembly methods for multiple joint modules can be deduced as:

$$N_{Com} = \sum_{i=1}^{N_{Var}} C_{N_{Var}}^{i} \cdot i! 2^{N_{Var}-1} \tag{3.11}$$

Where, N_{Var} is the total number of degrees of freedom of the manipulator, that is, the total number of joint modules under a specific assembly mode.

In the evaluation process of multi-scene modular series robot arm for different work scenarios and task requirements, the structure of the robot arm can be preprocessed from the two levels of system energy consumption and work space based on the optimization algorithm of redundant degrees of freedom robot arm according to specific requirements, and the fixed mechanism can be mapped to different mechanisms with different needs through modular reorganization The real-time optimization expands the space optimization capabilities of rigid robotic arms to meet diverse needs.

3.4.2 Kinematic Solution of Robotic Arm

For the redundant manipulator, in order to realize the motion trajectory planning of the end effector and the optimization of the degrees of freedom of each joint, it is necessary to carry out forward kinematics and inverse kinematics solutions according to the kinematic model of the corresponding assembled manipulator based on the homogeneous linkage transformation matrix in equation (3.10).

Knowing the joint angle θ_i of the manipulator, the kinematic model of the assembled form manipulator can be used to calculate the pose of the target coordinate system of the end effector of the robot relative to the base coordinate system, namely:

$$\mathrm{r}(t) = f(\theta_i(t))$$

(3.12)

Where $\mathrm{r(t)} \in \mathbb{R}^m$ is defined as the motion trajectory of the end-effector of A robotic arm, where m represents the dimension of the motion space and is defined as the spatial vector representation of the angles of each joint, while n represents the dimension of the joint space of the robotic arm and $f(\theta_i(t))$ represents the kinematics equation of the robotic arm. In redundant robotic arms, the dimension of the end-motion space m and the Angle dimension of each scattered joint n have the characteristics of $\mathrm{m} > \mathrm{n}$.

$$f(\theta_i(t)) = \mathrm{r}(t)$$

(3.13)

In A redundant robot arm, the end motion space dimension m and the Angle dimension n of each scattered joint have the characteristics of $\mathrm{m} > \mathrm{n}$. For a series robot arm, the solution of the forward kinematics is unique, while the solution of the inverse kinematics is not unique.

The general kinematic equation of series manipulator can be expressed as:

$$f(\theta_i) = \prod_{i=1}^{N_{Var}} T^{i-1}{}_i(\theta_i) = \begin{bmatrix} n_x & o_x & a_x & p_x \\ n_y & o_y & a_y & p_y \\ n_z & o_z & a_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(3.14)

Converted to scalar form, we get:

$$\mathrm{n}_{3\times1} = n(\theta) = n(\theta_1, \theta_2, \theta_3, \cdots, \theta_{N_{Var}})$$

(3.15)

$$\mathrm{o}_{3\times1} = o(\theta) = o(\theta_1, \theta_2, \theta_3, \cdots, \theta_{N_{Var}})$$

(3.16)

$$\mathrm{a}_{3\times1} = a(\theta) = a(\theta_1, \theta_2, \theta_3, \cdots, \theta_{N_{Var}})$$

(3. 17)

$$\mathrm{p}_{3\times1} = p(\theta) = p\left(\theta_1, \theta_2, \theta_3, \cdots, \theta_{N_{Var}}\right)$$

(3. 18)

It can be abbreviated as:

$$P_{12\times1} = f(\theta_i)$$

(3. 19)

Where $\theta_i \in \mathbb{R}^n$.

3.4.3 Robotic Arm Trajectory Planning

In order to avoid unnecessary vibration and impact caused by uneven motion, the motion trajectory function and its first and second derivatives (physical meaning are displacement, velocity, and acceleration respectively) are required to be smooth and continuous. The modular tandem manipulator is assembled by a plurality of joint modules in a specific way, and the trajectory planning of the joint space is to plan the smooth motion trajectory of each joint based on the joint motion constraints. In order to satisfy the constraints on multiple factors such as the angle, velocity, and acceleration of the joint movement, a fifth-order polynomial function is used:

$$\theta_i(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5$$

(3. 20)

The trajectory function $\theta_i(t)$ needs to meet the six constraints of the start and end point angles, velocity and acceleration, where the angle constraints:

$$\begin{cases} \theta(0) = \theta_0 \\ \theta(t_f) = \theta_f \end{cases}$$

(3.21)

The first derivative of the correspondence to implement the velocity constraint:

$$\begin{cases} \dot{\theta}(0) = \theta_0 \\ \dot{\theta}(t_f) = \theta_f \end{cases}$$

(3.22)

The second derivative of the correspondence to implement the acceleration constraint:

$$\begin{cases} \ddot{\theta}(0) = \theta_0 \\ \ddot{\theta}(t_f) = \theta_f \end{cases}$$

(3.23)

Substituting the constraints in (3.21), (3.22), and (3.23) into the (3.20) fifth-order polynomial function to obtain the system of equations:

$$\begin{cases} \theta_0 = a_0 \\ \theta_f = a_0 + a_1 t_f + a_2 t_f^2 + a_3 t_f^3 + a_4 t_f^4 + a_5 t_f^5 \\ \dot{\theta}_0 = a_1 \\ \dot{\theta}_f = a_1 + a_2 t_f + a_3 t_f^2 + a_4 t_f^3 + a_5 t_f^4 \\ \ddot{\theta}_0 = 2a_2 \\ \theta_f = 2a_2 + 6a_3 t_f + 12a_4 t_f^2 + 20a_5 t_f^3 \end{cases}$$

(3.24)

Solving Eq. (3.24), the coefficients of the fifth polynomial are:

$$\begin{cases} a_0 = \theta_0 \\ a_1 = \dot{\theta}_0 \\ a_2 = \dfrac{\ddot{\theta}}{2} \\ a_3 = \dfrac{20(\theta_f - \theta_0) - (8\dot{\theta}_f + 12\dot{\theta}_0)t_f - (3\ddot{\theta}_0 - \ddot{\theta}_f)t_f^2}{2t_f^3} \\ a_4 = \dfrac{30(\theta_0 - \theta_f) - (14\dot{\theta}_0 + 16\dot{\theta}_f)t_f - (3\ddot{\theta}_0 - 2\ddot{\theta}_f)t_f^2}{2t_f^4} \\ a_5 = \dfrac{12(\theta_f - \theta_0) - (6\dot{\theta}_f + 6\dot{\theta}_f)t_f - (\ddot{\theta}_0 - \ddot{\theta}_f)t_f^2}{2t_f^5} \end{cases}$$

(3.25)

By bringing the coefficient in (3.25) into the general formula of (3.20) of the fifth polynomial, the trajectory function $\theta_i(t)$ of two joint modules assembled in a certain direction can be obtained.

3.4.4 Kinematic Simulation of Robotic Arm

According to the above forward and inverse kinematics solution and the results of fifth-order polynomial trajectory planning, the kinematic simulation of the AyCxBy-D manipulator was carried out by using Robotics Toolbox for MATLAB.

Let the value of the initial joint variable of the end effector:

$$\text{InitialAngle} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

(3.26)

Let the end-effector terminate joint variable value:

$$\text{InitialAngle} = \begin{bmatrix} 0.7 \\ 1.2 \\ 0.15 \\ 2 \end{bmatrix}$$

(3.27)

As shown in Figure 3-23, you can obtain images of the angular displacement, velocity, and acceleration of Link 1 (Base-Ay linkage).

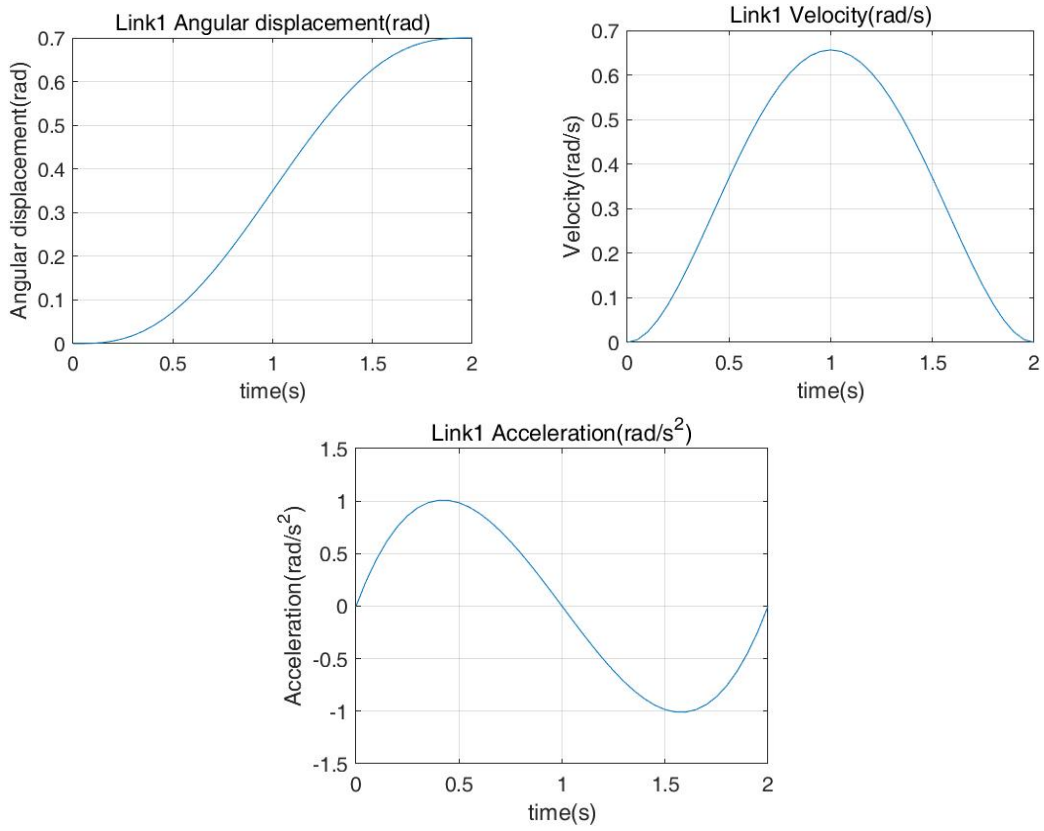


Figure 3-23 Link 1（Base-Ay）simulation data

As shown in Figure 3-24, the simulation data of Link 2 (Ay-Cx linkage).

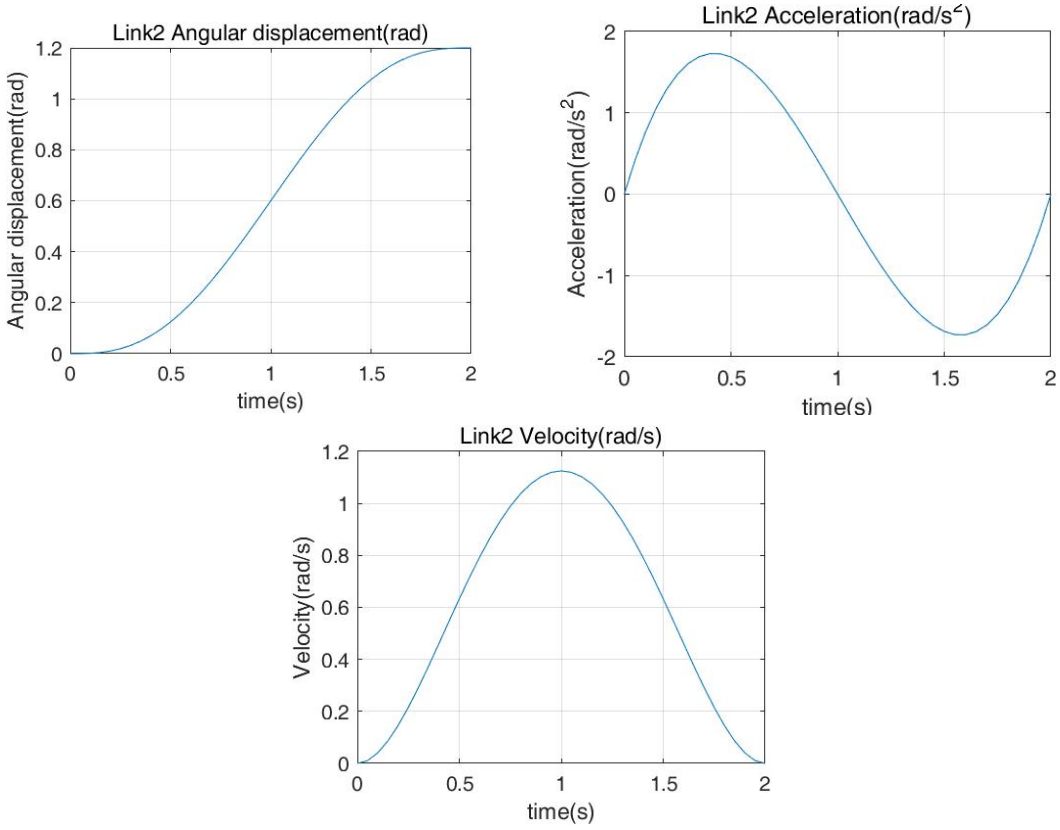


Figure 3-24 Link 2（Ay-Cx linkage）simulation data

As shown in Figure 3-25, the simulation data of Link 3 (Cx-By linkage).

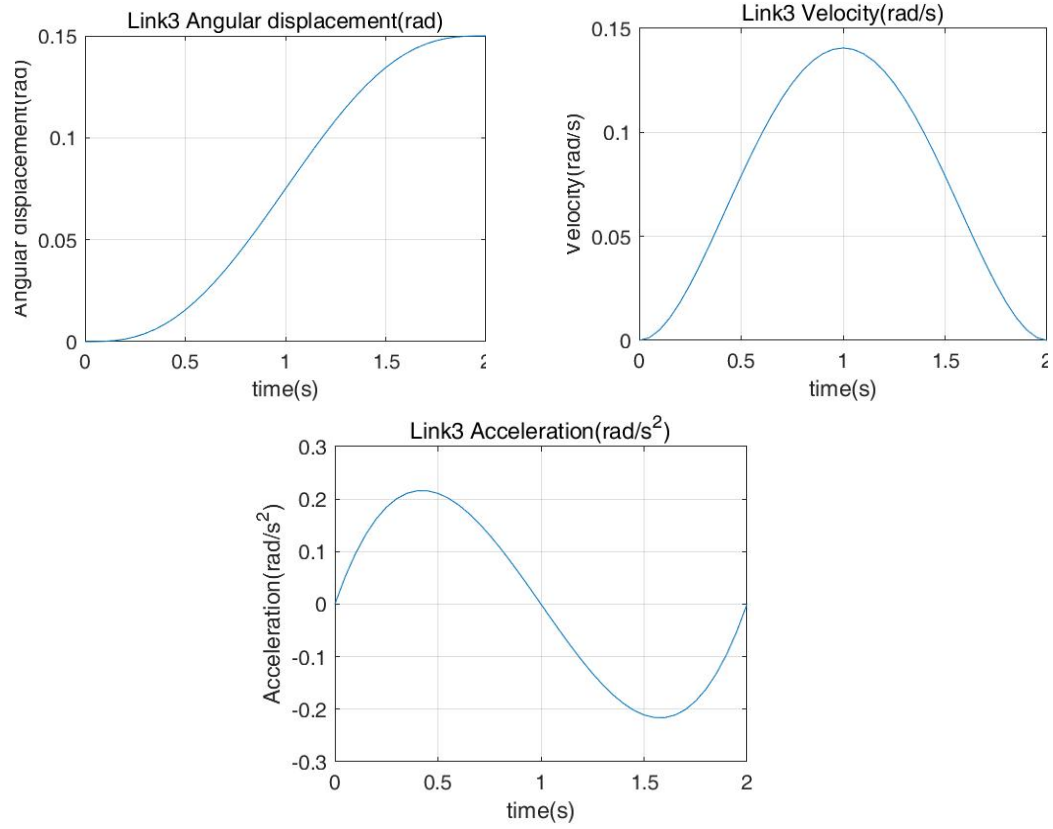


Figure 3-25 Link 3（Cx-By linkage）simulation data

As shown in Figure 3-26, the simulation data of Link 4 (By-D linkage).

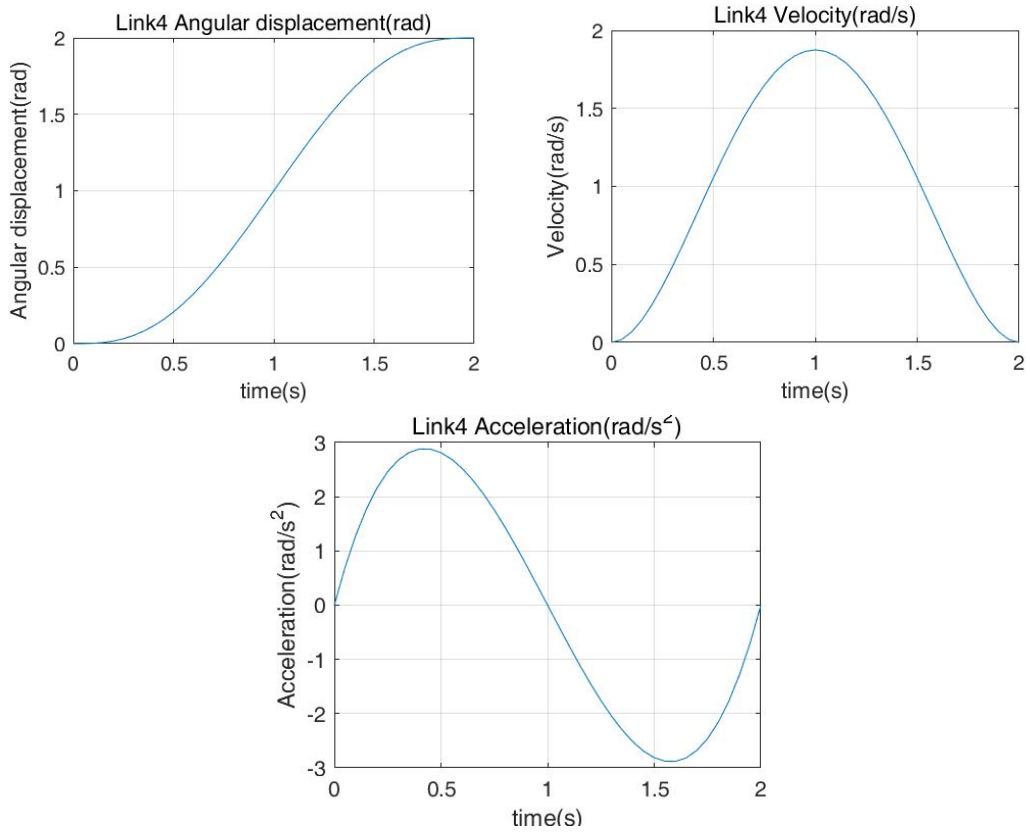


Figure 3-26 Link 3（By-D linkage）simulation data

As shown in Figure 3-27, the motion trajectory image can be obtained based on the calculation results of the spatial variables of the end effector.

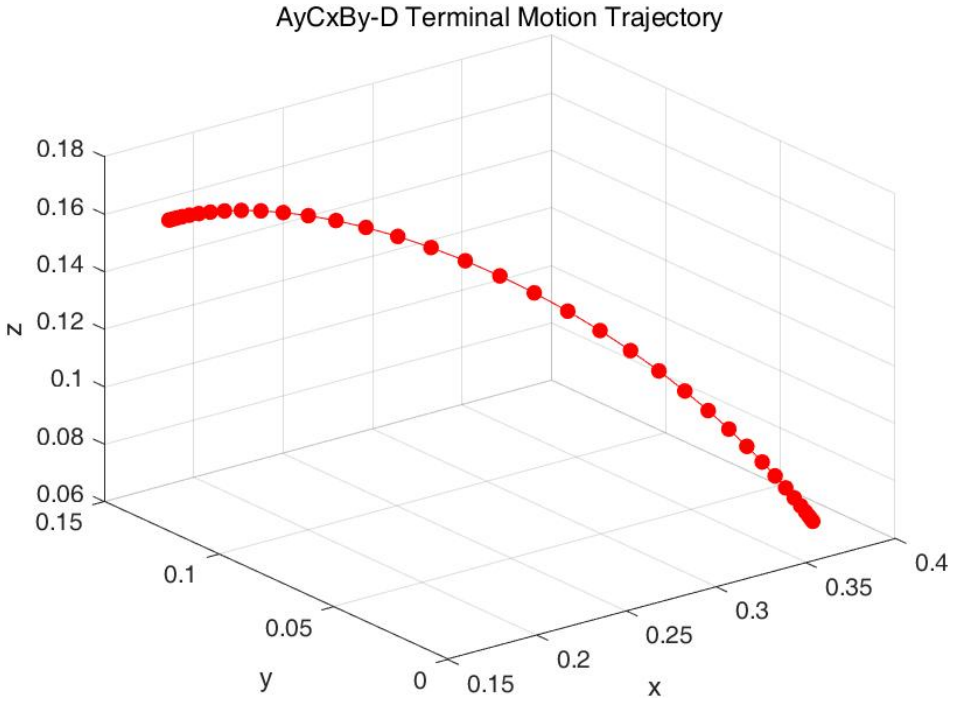


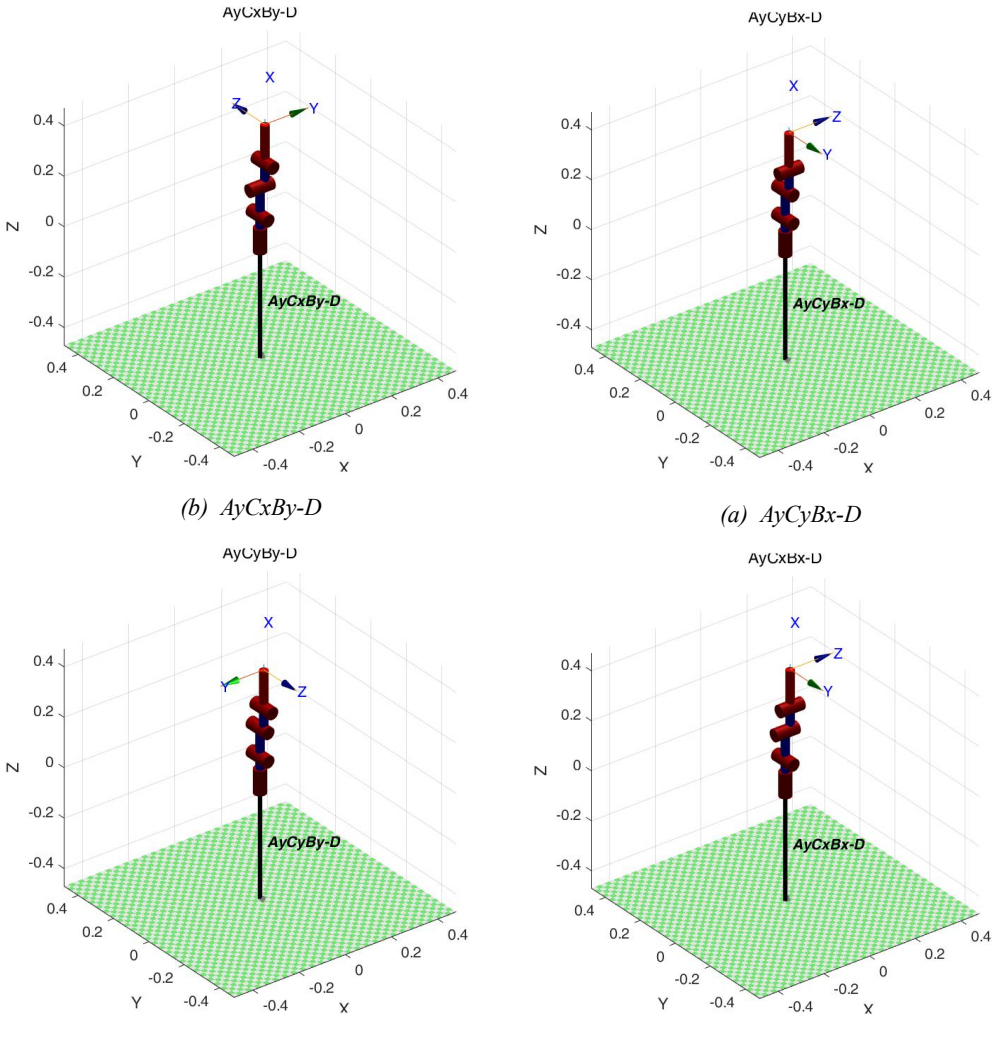
Figure 3-27 AyCxBy-D end-effector spatial motion trajectory

3.4.5 Optimization of Energy Consumption of Redundant Degrees of Freedom of the Robotic Arm

It is necessary to optimize the energy consumption of multiple drives to minimize the total energy required by the modular robotic arm system to complete a specific job by evaluating the total energy consumption of the system and pretreating the assembly method of the robotic arm.

The exhaustive method is used to integrate the second derivative (acceleration) of the corresponding operating angle of each joint module in all modular assembly structures that meet the task requirements, and the optimization index is designed:

$$E_{Sum} = \sum_{i=1}^{N_{Var}} \left| \int_0^t \ddot{\theta}_i(t) dt \right| \tag{3.28}$$



(b) AyCxBy-D (a) AyCyBx-D

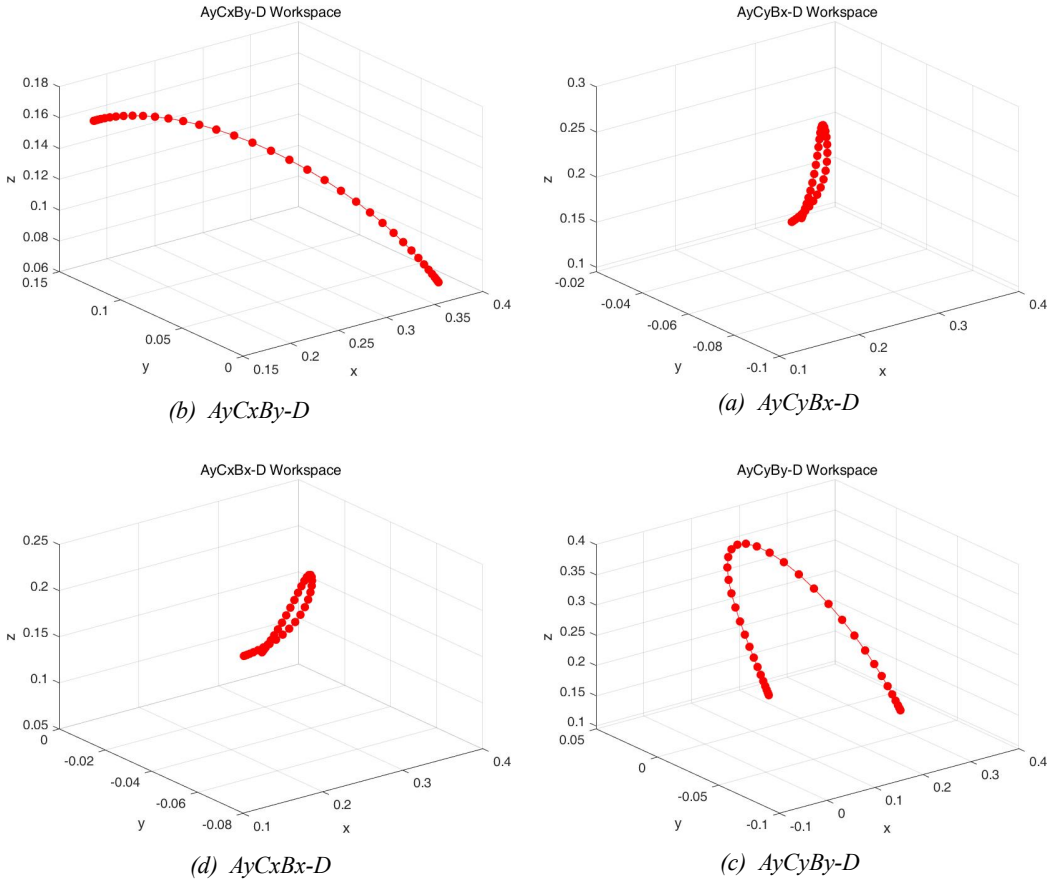
(d) AyCyBy-D (c) AyCxBx-D

Figure 3-28 ABC-D Robotic arm assembly model

3.4.6 Optimization of Redundant Degrees of Freedom Space of the Robotic Arm

In scenarios such as rescue and search and rescue, industrial automation production, and medical treatment, the total energy consumption requirements of the system are evaluated, and the working space of the robotic arm assembly method is preprocessed, so that the working space of multiple drives needs to be optimized to meet the specific task requirements when the modular robotic arm system completes specific tasks.

Using the exhaustive method, taking the ACB-D robotic arm as an example, the total number of modular assembly of the robotic arm is obtained according to formula (3.11), and the simulation model of each assembly mode is established separately by using the Robotics Toolbox for MATLAB tool, as shown in Figure 3-28.



(b) AyCxBy-D *(a) AyCyBx-D*

(d) AyCxBx-D *(c) AyCyBy-D*

Figure 3-29 ACB-D Robot arm end working space

The *qlim* attribute of the simulation model was used to obtain the range of motion of each joint module, and 10,000 random numbers were generated to define the variable matrix of each joint module. The random number is traversed, and the

random values in the range of motion are assigned to the variable matrix of each joint module to obtain enough variable values of each joint corresponding to the assembly mode of the robotic arm in a given range.

According to the test data in equations (3.26) and (3.27), the coordinate value of the end effector of the robotic arm is calculated. As shown in Figure 3-29, a scatter plot in a three-dimensional coordinate system is generated to visualize the working space of the end effector of the robotic arm with the corresponding assembly method.

Based on the obtained working space image, the modular assembly method that meets the task requirements is selected to reassemble the robot arm, optimize the redundant spatial freedom in real time, and greatly improve the spatial flexibility of the modular series robot arm.

4. Functional Verification of Multi-scene Modular Series Manipulator

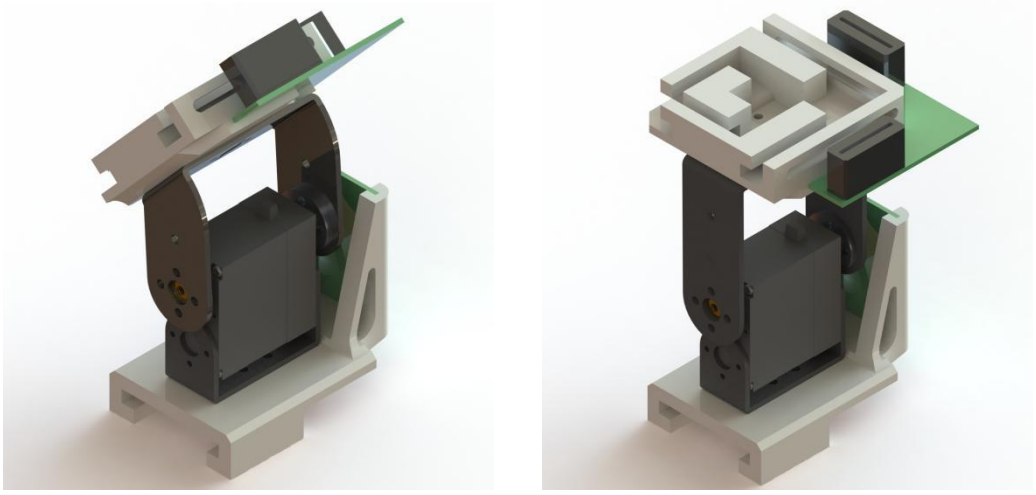
4.1 Overview of Basic Functional Verification

According to the preliminary conclusion of the structural design and analysis and optimization of the robotic arm, one base, one 45° joint module, two 90° joint modules, and three different end effectors were selected to make the prototype to verify the theoretical analysis results. The main mechanical structure of the robotic arm was 3D printed using light-curing technology, and the drive device and PCB circuit board were assembled to verify the feasibility of the multi-channel modular assembly strategy. The kinematic solution algorithm was migrated to the software debugging platform, and the redundant degree of freedom optimization method was used to preprocess the assembly structure of the robotic arm for specific task scenarios, and the feasibility of optimizing the application scope and work efficiency of the robotic arm through modular reorganization was verified under the transformation of different scenarios.

4.2 Structural Design Verification

4.2.1 Functional Verification of Modular Connection Components

As shown in Figure 4-1,Smooth the surface of each connection channel of the T connection node and N connection node made of photosensitive resin. Smooth and stable connection between the joint module of each module of the assembly drive device and the base and end effector. In order to ensure the stable fixing of the base during the operation of the manipulator, four latex suction cups are installed to make it fully fixed with the table top.



(b) 45 ° Joint Module (a) 90° Joint Module

Figure 4-1 Rendering of the joint module assembly

4.2.2 Hardware Structure Installation and Testing

EDA software is used to design PCB circuit boards, and electronic components are soldered to interface with signal terminals after proofing and production. Clean the surface of the PCB and power on the test, burn the test program, and each part can run stably as scheduled.

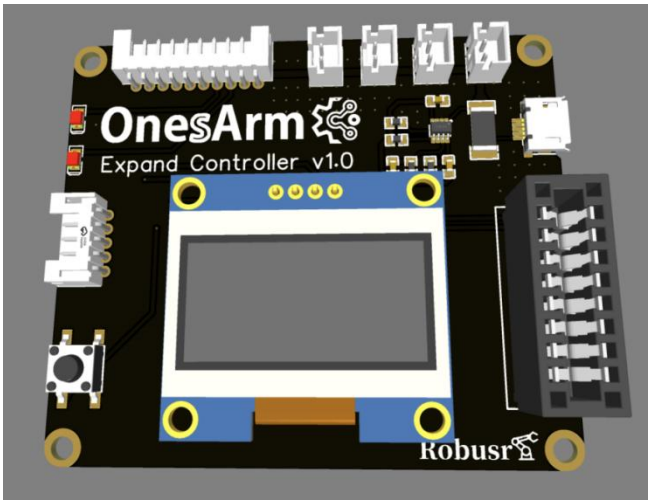


Figure 4-2 Extended module PCB model diagram

As shown in Figure 4-2, the test expansion module is designed, prototype and produced. It is connected to the main board of the base through the PH2.0 10P terminal line. The expansion module is equipped with a GNSS positioning module, which can return the current altitude pressure and coordinate information of the robot arm for special disaster rescue scenarios. A 0.96 "monochrome OLED screen to display the current detected data and the status of the real-time feedback from the

robot arm; A row of 82P gold finger connectors, can be connected to the joint module PCB sliding Mosaic components, can quickly debug each module; A USB TypeB interface for connecting external power supply resources when the robot arm is heavily loaded.

4.2.3 Construction and Debugging of Software Platform

A software platform was built based on MicroPython development tools, and a simple open-loop control was carried out on the robot arm test prototype to test the functions of various LED signal lights and signal interfaces, such as 0.96 inch OLED screen to display GNSS positioning data output and goldfinger socket debugging.

4.3 Analysis, Optimization and Verification

4.3.1 Kinematic Solution and Trajectory Planning Entity Verification

The kinematic solution algorithm is migrated to the MicroPython development platform, the test program is written and burned to the base controller of the robotic arm, and when the robotic arm is assembled in the form of AyCxBy-D, the initial point and the end point are defined, and the robotic arm can run according to a predetermined smooth trajectory.

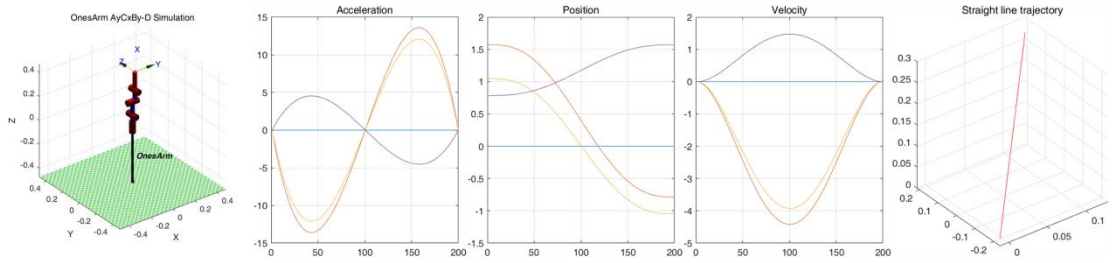


Figure 4-3 Verification of Robotic Arm Kinematics Solution

4.3.2 Verification of Redundant Degrees of Freedom Space Optimization Entities

The spatial motion trajectories of the end effector of the manipulator with each assembly mode under the constraints of specific starting point and ending point are output respectively. According to the specific space usage requirements, the corresponding spatial assembly mode is selected to modular reassemble the manipulator, achieving the purpose of optimizing the redundant freedom space of the manipulator.

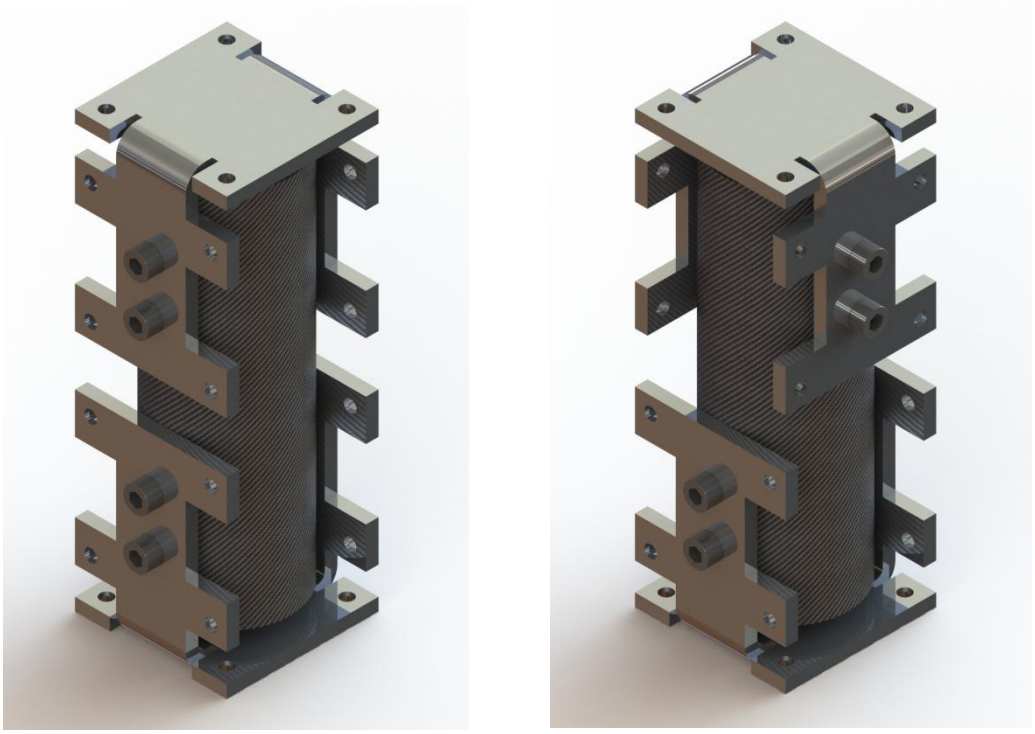
5. Prototype Experimental Platform and Analysis Method Update and Iteration

5.1 Follow-up Research and Update Iterative Overview

The first four chapters mainly contain the research progress as of mid-April 2024, and in the continuous learning and practice, the structure and analysis methods of the robotic arm have been improved, the modular node mechanism and analysis strategies have been optimized, and the prototype experimental platform has been updated and iterated. This is reviewed in this chapter.

5.2 Structural Design

5.2.1 Modular Linkages



(a) X-X Long Link *(b) X-Y Long Kink*

Figure 5-1 Rendering of a modular long-link model

As shown in Figure 5-1 and 5-2, in order to enrich the diversity of modular connection methods, long connecting rods and integrated short connecting rods in the

X-X AND X-Y directions are designed, and each standardized connection node is connected by M5 screws, and the overall 5052 aluminum alloy sheet metal and carbon fiber tube structure with a thickness of 5mm is used to ensure the strength and rigidity of the modular manipulator arm and realize the flexible connection of various structures. As shown in Figure 5-3, the static stress finite element analysis simulation is performed on the key stress nodes.

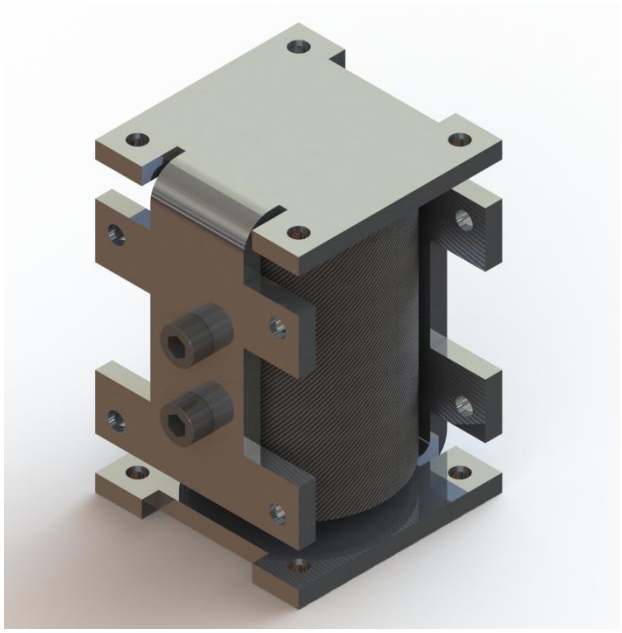


Figure 5-2 Rendering of a modular long-link model

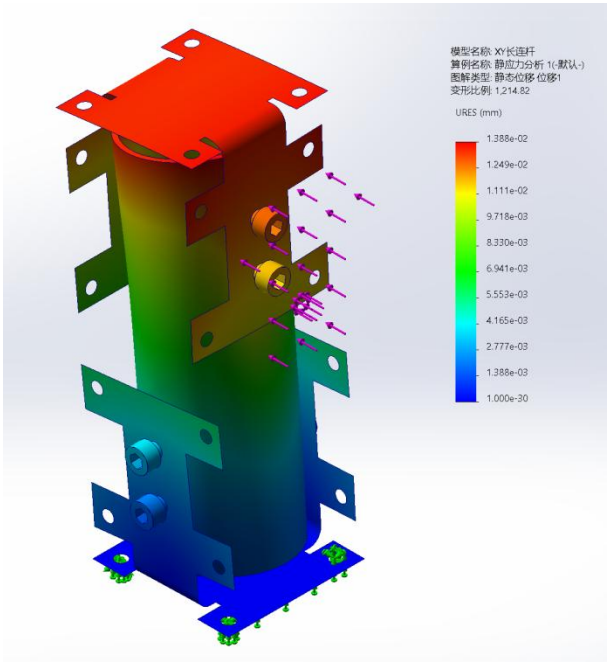


Figure 5-3 Simulation contour diagram of static stress finite element analysis of key stressed nodes

5.2.2 Modular Joints

As shown in Figure 5-4, the 42BYGH47 stepper motor and the FLE-42-20 planetary reducer are selected as the driving device, and the reduction ratio is 1:20. Equipped with a magnetically encoded closed-loop controller, a STM32F103C8T6 platform CAN protocol communication component and a standardized interface for the connection of each module, the stable control and accurate output torque of the modular robotic arm are realized.

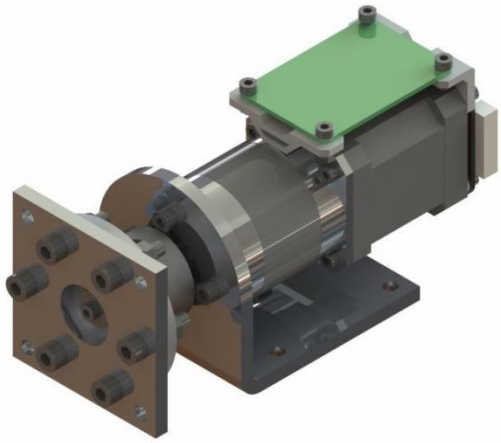


Figure 5-4 Rendering of the modular joint model

5.2.3 Working Platform

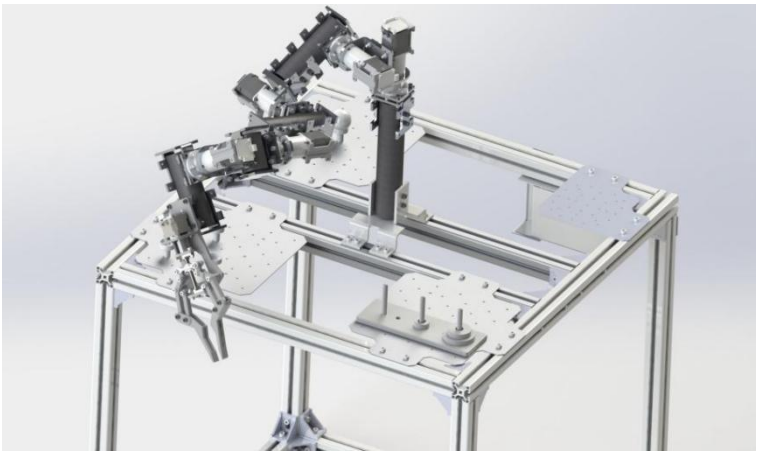
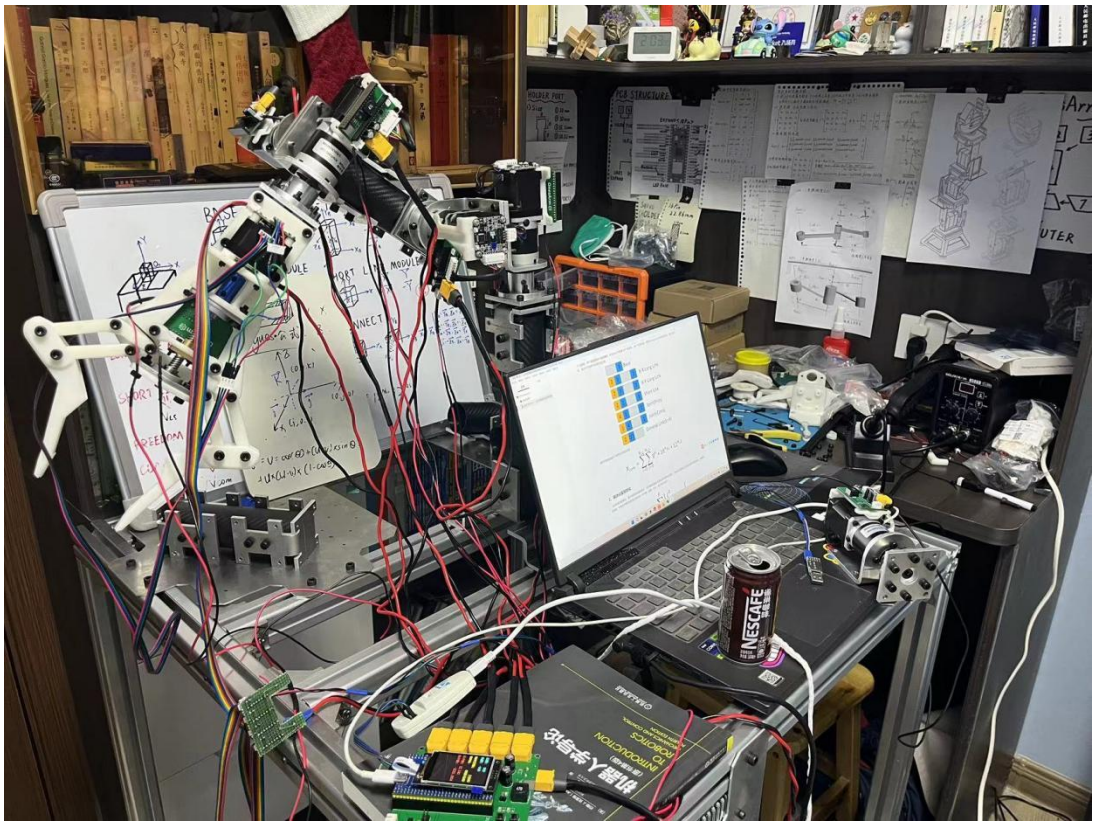


Figure 5-5 Working platform and robot arm overall structure model rendering

As shown in Figure 5-5, the working platform of the modular robotic arm is built with 4040 European standard aluminum profiles, and a 350W stable power input, 3 test positioning boards and a CAN bus controller of the STM32F407ZGT6 platform are installed, and the communication interface of the UART protocol is reserved, which is selected and installed according to the specific working scenarios and task requirements. As shown in Figure 5-6, the prototype platform is assembled and tested.



(a) Gripper actuators



(b) 16mm diameter clamp actuator

Figure 5-6 Prototype experimental platform assembly test

5.3 Analysis and Optimization

5.3.1 Modular Combination Analysis

As shown in Figure 5-7, according to the tandem structure of the manipulator, a rotating pair of surface contact is formed between the connecting rod module and the joint module; The connecting rod module and the connecting rod module also form a solid connection with the surface contact, so it can be unified into an "FCF structure" of connecting surfaces, connecting channels, and connecting surfaces.

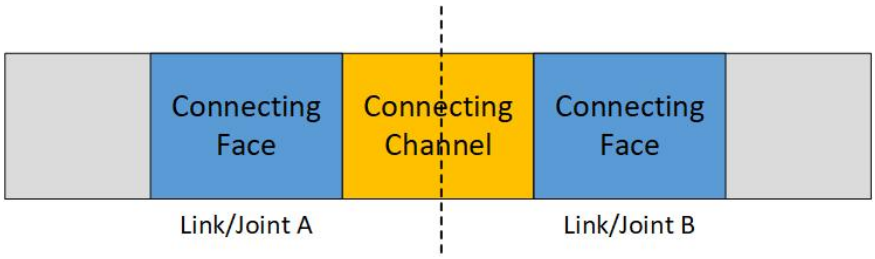


Figure 5-7 The basic pattern of FCF structure

Each module of the robotic arm is fixed from left to right, and the overall series structure can be arranged into an abstract model in Figure 5-8.

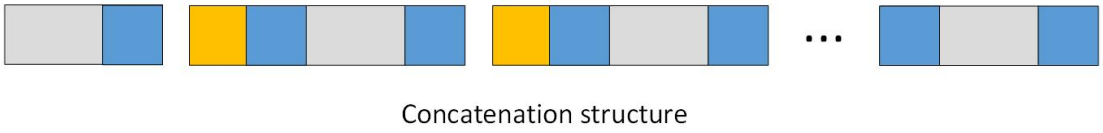


Figure 5-8 FCF structure of tandem robotic arm

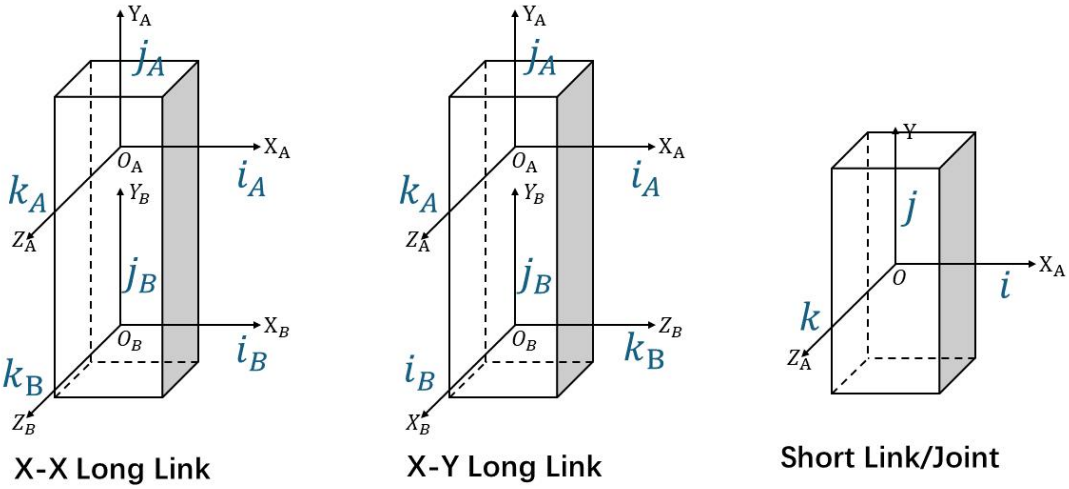


Figure 5-9 Simplified model of each modular node

As shown in Figure 5-9, the coordinate system is established perpendicular to

each connecting surface of the connecting surface of the connecting rod module and the joint module as the coordinate origin.

Among them, X-X Long Link and X-Y Long Link can be abstracted into 6 connecting surfaces, each with 4 connecting channels, Short Link can be abstracted into 4 connecting surfaces, each connecting surface has 4 connecting channels, and Joints can be abstracted into 2 connecting surfaces with 4 connecting channels. According to the Rodriguez formula and assembly relationship, the coincidence items are deleted, as shown in Figure 5-10, and the total number of methods corresponding to the "FCF structure" is obtained.

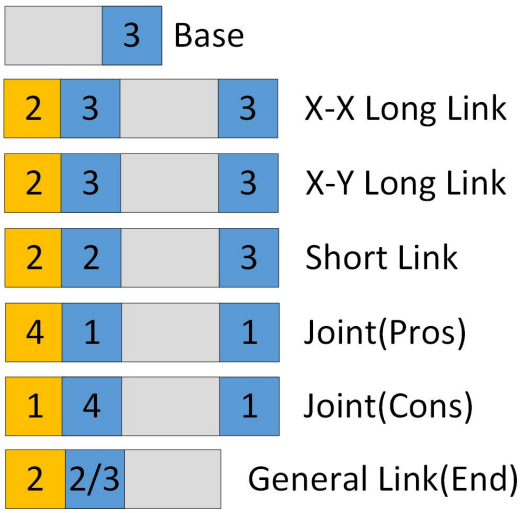


Figure 5-10 Total number of FCF connection methods for each modular node

The total number of modular combinations with different degrees of freedom (F) is obtained:

$$N_{Com} = \sum_{i=0}^{N_{LL}} \sum_{j=0}^{N_{LS}} 8^{F} * 18^{N_{LL}} * 12^{N_{LS}}$$

5.3.2 Modular Reorganization and Optimization

In a variety of application scenarios, it is necessary to evaluate the total energy consumption demand of the system, preprocess and modularize the assembly mode of the robotic arm to minimize the total power consumption required by the system to complete a specific job, and integrate the first derivative (velocity) of the corresponding operating angle of each joint module according to the trajectory

function, and design the optimization index:

$$E_{Sum} = \sum_{i=1}^{N_{Var}} \left| \int_0^t \ddot{\theta}_i(t) dt \right|$$

At the same time of power consumption optimization, the working space of the robotic arm needs to be optimized to meet the optimal conditions when completing the target task, and the simulation model is established according to the DH table of the specific assembly method, and the motion trajectory of the test point is output. Figure 5-11 shows the basic process of completing a specific job.

6. Conclusions and Prospects

In the process of this research, drawing on the advantages of continuous manipulators, improving on the basis of traditional rigid manipulators, the modular connection components were designed according to the mortise and tenon structure, the structural design of the multi-scene modular tandem manipulator was completed, and the static stress finite element analysis and preliminary functional verification based on SolidWorks were used to evaluate the feasibility of the design. The spatial characteristics of the modular combination of each joint of the manipulator were analyzed by permutation, combination and enumeration method were used, the kinematic model of the manipulator was constructed based on the coordinate transformation method, the forward and reverse kinematics solution was completed by MATLAB, and the single and multi-joint of the manipulator were simulated and analyzed, and the accurate images of the pose, velocity and acceleration of the end effector were obtained. A multi-scene switching and modular servo control system is built, which can accurately control the robotic arm according to the kinematic calculation and simulation analysis results, ensure the stability and reliability of performing specific work in different environments, greatly improve the flexibility and versatility of the modular robotic arm, and realize the purpose of easy maintenance and expansion of the robotic arm and space optimization.

Simulation is an indispensable part of the research field of robotics, and it is one of the necessary approaches to study the joint characteristics and performance of robots with the help of simulation [8]. In the follow-up research, the modular structure of the manipulator should be optimized according to the results of static stress finite element analysis, the dynamic equation should be established, the dynamic mode analysis of the manipulator should be completed, and the PID algorithm should be introduced to improve the control system of the manipulator.

With the development of the Internet of Things and cloud computing technology, the modular tandem robotic arm can realize remote monitoring and management,

improving production efficiency and response speed. By combining biomimicry [7], artificial intelligence, and machine learning algorithms.

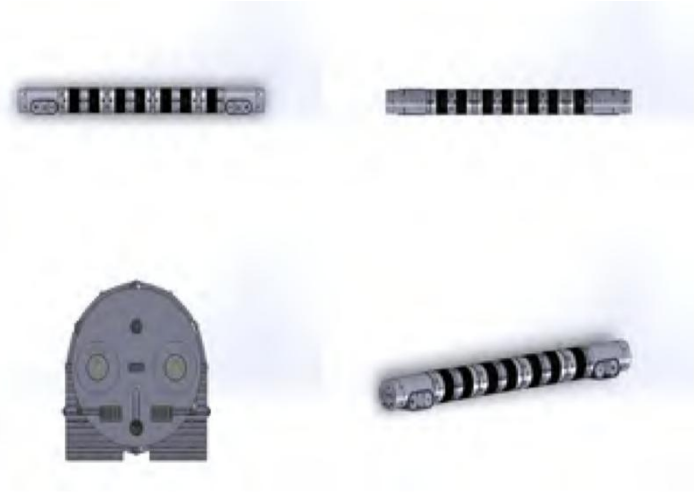


Figure 6-1 Bionic serpentine search and rescue robot based on modular technology [9]

Drawing on the wisdom of nature, the robotic arm is updated and iterated with a more ingenious structure, completes the functions of target recognition and intelligent analysis of autonomous space optimization, better adapts to the task requirements in different scenarios, improves work efficiency and execution accuracy, and achieves the purpose of multi-scene modular robotic arm bead and arm combination.

My hometown, Ningxia, is known as the "hometown of wolfberries", and with the rapid development of technology, automated wolfberry picking is facing a huge challenge. The traditional vibration method has not been widely implemented due to its high loss and low efficiency. In the follow-up exploration process, the multi-scene modular robotic arm studied in this topic optimizes the structure and algorithm, pretreats the structure of the robotic arm according to different wolfberry picking needs, flexibly responds to a variety of scenarios according to the corresponding varieties and tree shape modular reorganization, and adds machine vision components to realize the accurate identification and separation of flowers and fruits, improve the production efficiency of 10,000 mu of wolfberry fruit fields in his hometown, and contribute to the construction of beautiful Ningxia Sichuan. Let the high-tech industry be integrated into grassroots production, and intelligent machinery and sweet wolfberry will be combined!

Acknowledgements

During the research process of this project, I have experienced many challenges, but I have been fortunate to receive support and assistance from numerous individuals. Here, I would like to express my sincerest gratitude to everyone who has cared for and supported me.

I am especially grateful to my parents. Although they are not professionals in the relevant field, they have provided substantial financial and equipment support over the years, transforming my living space into a mechanical and electronic workshop where I can design, assemble, and test my work, despite the lack of external laboratory resources. Their immense understanding and comfort during my demanding studies and research have made them the most solid backing on my path of technological innovation. I wish my parents good health and success in their work, and I hope to repay their earnest expectations soon.

I would also like to thank Teacher Ma Bowei from the school's Robotics Innovation Studio for providing valuable resources, including topic analysis, literature review channels, information on related competitions, and guidance on paper formatting. He accompanied me to various levels of technology innovation competitions, offering encouragement, care, and support in his spare time. He opened up a new platform for me, allowing me to pursue my passions with determination.

I am grateful to Senior Yang Xu from the University of Electronic Science and Technology, who answered my questions during my self-study of relevant professional knowledge and the writing of my thesis. He provided invaluable advice on motor selection for robotic arms, overall structural design, and load testing. His passion for robotics has deeply inspired me and motivated me to follow in his footsteps on the path of learning and research.

I would like to thank Associate Professor Shao Jinlong from the College of Mechanical Engineering at Ningxia University. Taking the opportunity of participating in the 38th National Youth Science and Technology Innovation Competition, he provided professional guidance for my project under the arrangement

of the provincial science and technology association. He acknowledged the successful aspects of my bold innovations while pointing out the deficiencies in the mechanical structure's mechanical performance, encouraging me to remain true to my original aspirations and strive for progress in my future studies and life. I greatly admire his rigorous academic attitude and approachable personality.

The preliminary completion of this project represents an important exploration and breakthrough in my lifelong passion for electronics and mechanics. With a grateful heart, I will continue to study diligently and solidify my professional knowledge foundation. I will not forget my original intentions and will keep moving forward. Finally, I would like to express my heartfelt thanks to all the experts and judges who took time out of their busy schedules to review this thesis.

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Appendix

Research Results Obtained

[1] At the beginning of 2022, an indoor intelligent management system based on an embedded platform was designed for real-time monitoring of indoor environmental conditions, visual management of space personnel, and voice interaction. The work won the second prize of Guyuan Science and Technology Innovation Competition.

[2] At the beginning of 2023, a modular telemetry system for meteorological detection rockets was designed, which organized the structure of sandstorm detection rockets in a modular way, expanded the detection types and scope through standardized interfaces of sensors, and realized real-time analysis and accurate recording of rocket modular control and data in a concise and efficient manner. In view of the sand and dust weather, diversified meteorological detection needs have been realized. The work won the first prize of the 37th Ningxia Youth Science and Technology Innovation Competition and the second prize of the 37th National Youth Science and Technology Innovation Competition.

[3] At the end of 2023, he was invited to participate in the recording of Shanghai TV's "Young Edison: Tomorrowland".

[4] The project won the first prize in the 37th Ningxia Youth Science and Technology Innovation Competition, and the first prize in the 38th National Youth Science and Technology Innovation Competition in July 2024